

SingleRAN

# EPC-based NSA Experience Improvement Feature Parameter Description

|       |            |
|-------|------------|
| Issue | Draft B    |
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## **Huawei Technologies Co., Ltd.**

Address: Huawei Industrial Base  
Bantian, Longgang  
Shenzhen 518129  
People's Republic of China

Website: <https://www.huawei.com>

Email: [support@huawei.com](mailto:support@huawei.com)

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# 1 Change History

The following describes the specific changes in different issues.

## SRAN22.1 Draft B (2026-01-30)

This issue includes the following changes.

- Technical changes  
None
- Editorial changes
  - Added the description of an NSA UE meeting the conditions for triggering an SCC addition procedure. For details, see [3.1 Principles](#).

## SRAN22.1 Draft A (2025-12-31)

This issue introduces the following changes to SRAN21.1 06 (2025-11-07).

- Technical changes

| Change Description                                                                                                                                   | Parameter Change | Base Station Model                                                              |
|------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|---------------------------------------------------------------------------------|
| Added incompatibility of BTS5929R with NSA carrier combination selection based on user experience. For details, see <a href="#">3.3.3 Hardware</a> . | None             | 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R |
| Added incompatibility of BTS5929R with NSA anchor coverage extension based on user experience. For details, see <a href="#">4.3.3 Hardware</a> .     | None             | 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R |

| Change Description                                                                                                                                                                                                                                                                                                                                                                              | Parameter Change                                                                                                                                                                                                                                                                                                                                                                                        | Base Station Model                                                                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Added the timer specifying the waiting time for stopping measurement after all inter-frequency measurement reports are received. For details, see:</p> <ul style="list-style-type: none"> <li>• <a href="#">3.1 Principles</a></li> <li>• <a href="#">3.2.2 Impacts</a></li> <li>• <a href="#">3.4.2.1 Data Preparation</a></li> <li>• <a href="#">3.4.2.2 Using MML Commands</a></li> </ul> | <p>Added the <b>CellMultiCarrUniSch.StopMeasAfterInterFMrTmr</b> parameter on the LTE side.</p>                                                                                                                                                                                                                                                                                                         | <ul style="list-style-type: none"> <li>• 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R</li> <li>• DBS3900 LampSite and DBS5900 LampSite</li> </ul> |
| <p>Added the reserved parameters for "UE-level capability negotiation for NSA carrier combination selection based on user experience" to the disuse list. For details, see:</p> <ul style="list-style-type: none"> <li>• <a href="#">3.4.2.1 Data Preparation</a></li> <li>• <a href="#">3.4.2.2 Using MML Commands</a></li> </ul>                                                              | <p>Modified parameter on the LTE side: Added the <b>BIT8</b> option of the <b>eNBRsvdPara.RsvdSwPara5</b> parameter to the disuse list. The controlled function takes effect by default.</p> <p>Modified parameter on the NR side: Added the <b>RSVDSWPARAM0_BIT1</b> option of the <b>gNBRsvdExt01.RSVDSWPARAM0</b> parameter to the disuse list. The controlled function takes effect by default.</p> | <ul style="list-style-type: none"> <li>• 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R</li> <li>• DBS3900 LampSite and DBS5900 LampSite</li> </ul> |

| Change Description                                                                                                                                                                                                                                                                                                                                                                                                                                                | Parameter Change                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Base Station Model                                                                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Optimized the function "NSA carrier combination selection based on user experience". For details, see:</p> <ul style="list-style-type: none"> <li>• <a href="#">3.1 Principles</a></li> <li>• <a href="#">3.2.2 Impacts</a></li> <li>• <a href="#">3.3.2.1 Prerequisite Functions</a></li> <li>• <a href="#">3.4.2.1 Data Preparation</a></li> <li>• <a href="#">3.4.2.2 Using MML Commands</a></li> <li>• <a href="#">3.4.4 Network Monitoring</a></li> </ul> | <p>Added parameters on the LTE side:</p> <ul style="list-style-type: none"> <li>• <b>NsaDcMgmtConfig.NsaMinDLAvgToBeSchedUeNum</b></li> <li>• <b>NsaDcAlgoParam.NsaGFactorThldForHoTrgCell</b></li> </ul> <p>Modified parameters on the LTE side:</p> <ul style="list-style-type: none"> <li>• Added the <b>NSA_NR_SER_FREQ_B1_MEAS_OPT_SW</b> option to the <b>EnodebAlgoExtSwitch.MultiNetworkingOptionOptSw</b> parameter.</li> <li>• Added the <b>NSA_NO_NR_B1_BLI_ND_ADD_SCG_SW</b> option to the <b>EnodebAlgoExtSwitch.NsaDcAlgoExtSwitch</b> parameter.</li> <li>• Added the <b>NSA_NR_STRONGEST_PCELL_EVAL_SW</b> option to the <b>EnodebAlgoExtSwitch.NsaDcAlgoExtSwitch</b> parameter.</li> <li>• Added the <b>NSA_NR_SCC_GRID_EVAL_OPT_SW</b> option to the <b>EnodebAlgoExtSwitch.NsaDcAlgoExtSwitch</b> parameter.</li> <li>• Added the <b>NSA_SEL_SCG_FAIL_VG_PUNISH_SW</b> option to the <b>EnodebAlgoExtSwitch.NsaDcAlgoExtSwitch</b> parameter.</li> </ul> | <ul style="list-style-type: none"> <li>• 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R</li> <li>• DBS3900 LampSite and DBS5900 LampSite</li> </ul> |



| Change Description                                                                                                                                                                                                                                                                                                                                                                                                                             | Parameter Change                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Base Station Model                                                                                                                                                               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                | Modified parameter on the NR side: Added the <b>NSA_ACTUAL_CFG_SCC_EVAL_SW</b> option to the <b>gNodeBParam.Ne<sup>work</sup>kingOptionOptSw</b> parameter.                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                  |
| Added the function "optimized handling of procedure conflicts upon RRC connection reestablishments of NSA UEs". For details, see <a href="#">3.2.2 Impacts</a> .                                                                                                                                                                                                                                                                               | Modified parameter on the LTE side: Added the <b>NSA_REEST_FLOW_CONFLICT_OPT_SW</b> option to the <b>EnodeBAlgoExtSwitch.NsaDcAlgoExtSwitch</b> parameter.                                                                                                                                                                                                                                                                                                                                                                                                                                  | <ul style="list-style-type: none"> <li>3900 and 5900 series base stations (macro base stations) excluding the BTS5929R</li> <li>DBS3900 LampSite and DBS5900 LampSite</li> </ul> |
| Added support for NSA carrier combination selection triggered by coverage-based NR SCC release. For details, see: <ul style="list-style-type: none"> <li><a href="#">NSA Carrier Combination Selection Based on User Experience</a> in <a href="#">3.1 Principles</a></li> <li><a href="#">3.3.2.1 Prerequisite Functions</a></li> <li><a href="#">3.4.2.1 Data Preparation</a></li> <li><a href="#">3.4.2.2 Using MML Commands</a></li> </ul> | Modified parameter on the NR side: Added the <b>NR_SCC_REL_NSACARR_SEL_SW</b> option to the <b>NRCellCaMgmtConfig.CaStrategySwitch</b> parameter.<br>Modified parameters on the LTE side: <ul style="list-style-type: none"> <li>Added the <b>NR_SCC_REL_NSACARR_SEL_SW</b> option to the <b>NsaDcMgmtConfig.NsaDcAlgoEnhSwitch</b> parameter.</li> <li>Added the <b>NR_SCC_REL_NSACARR_SEL_FLAG</b> option to the <b>NrScgFreqConfig.AggregationAttribute</b> parameter.</li> </ul> Added parameter on the LTE side: Added the <b>NsaDcMgmtConfig.NrScRelCarrSelPunishTimer</b> parameter. | <ul style="list-style-type: none"> <li>3900 and 5900 series base stations (macro base stations) excluding the BTS5929R</li> <li>DBS3900 LampSite and DBS5900 LampSite</li> </ul> |

| Change Description                                                                                                                                                                                                                                                                                                                                                                  | Parameter Change                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Base Station Model                                                                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Replaced the reserved parameters for "optimized experience evaluation for NSA carrier combination selection based on user experience" with formal parameters. For details, see:</p> <ul style="list-style-type: none"> <li>• <a href="#">3.1 Principles</a></li> <li>• <a href="#">3.4.2.1 Data Preparation</a></li> <li>• <a href="#">3.4.2.2 Using MML Commands</a></li> </ul> | <p>Modified parameters on the LTE side: Added the <b>RSVD_SW_PARAM0_BIT20</b> option of the <b>EnodebRsvdParamExt.RsvdSwParam0</b> parameter to the disuse list and replaced it with the <b>NSA_NR_EVAL_REQ_N_R_MEAS_IND_SW</b> option of the <b>EnodebAlgoExtSwitch.NsaDcAlgoExtSwitch</b> parameter.</p> <p>Modified parameters on the NR side: Added the <b>RSVDSWPARAM4_BIT19</b> option of the <b>gNBRsvd.RsvdSwParam4</b> parameter to the disuse list and replaced it with the <b>EXP_BASED_NSA_PCC_SEL_OPT_SW</b> option of the <b>gNodeBAlgo.NrSmartMultiCarrSw</b> parameter.</p> | <ul style="list-style-type: none"> <li>• 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R</li> <li>• DBS3900 LampSite and DBS5900 LampSite</li> </ul> |
| <p>Added the function "non-gap-assisted B1 measurement of NR SCG frequencies for NSA DC UEs". For details, see:</p> <ul style="list-style-type: none"> <li>• <a href="#">3.1 Principles</a></li> <li>• <a href="#">3.2.2 Impacts</a></li> <li>• <a href="#">3.4.2.1 Data Preparation</a></li> <li>• <a href="#">3.4.2.2 Using MML Commands</a></li> </ul>                           | <p>Modified parameters on the LTE side:</p> <ul style="list-style-type: none"> <li>• Added the <b>NSA_DC_NR_SCG_B1_NO_GAP_SW</b> option to the <b>EnodebAlgoExtSwitch.NsaDcAlgoExtSwitch</b> parameter.</li> <li>• Added the <b>NSA_DC_NR_SCG_FB_D_B1_NO_GAP_SW</b> option to the <b>UeCompat.BlacklistControlExtSwitch2</b> parameter.</li> </ul>                                                                                                                                                                                                                                          | <ul style="list-style-type: none"> <li>• 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R</li> <li>• DBS3900 LampSite and DBS5900 LampSite</li> </ul> |

- Editorial changes

- Added configuration descriptions of some optional MML commands. For details, see [3.4.2.2 Using MML Commands](#).

# 2 About This Document

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## 2.1 General Statements

### Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

#### NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

### Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in this document apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

## 2.2 Applicable RAT

This document applies to LTE FDD, LTE TDD, and NR.

For definitions of base stations described in this document, see section "Base Station Products" in *SRAN Networking and Evolution Overview*.

## 2.3 Features in This Document

This document describes the following features.

| RAT     | Feature ID        | Feature Name                                                         | Chapter/Section                                                              |
|---------|-------------------|----------------------------------------------------------------------|------------------------------------------------------------------------------|
| LTE FDD | MRFD-171<br>223   | NSA Carrier Combination Selection Based on User Experience (LTE FDD) | <a href="#">3 NSA Carrier Combination Selection Based on User Experience</a> |
| LTE TDD | MRFD-171<br>233   | NSA Carrier Combination Selection Based on User Experience (LTE TDD) |                                                                              |
| NR      | MRFD-171<br>263   | NSA Carrier Combination Selection Based on User Experience (NR)      |                                                                              |
| LTE FDD | LNOFD-181<br>303  | NSA Anchor Coverage Extension Based on User Experience               | <a href="#">4 NSA Anchor Coverage Extension Based on User Experience</a>     |
| LTE TDD | TDLOFD-18<br>1210 | NSA Anchor Coverage Extension Based on User Experience               |                                                                              |

## 2.4 Differences

### Differences Between LTE FDD and LTE TDD

| Function Name                                              | Difference                                                                                                                                                                           | Chapter/Section                                                              |
|------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| NSA carrier combination selection based on user experience | None                                                                                                                                                                                 | <a href="#">3 NSA Carrier Combination Selection Based on User Experience</a> |
| NSA anchor coverage extension based on user experience     | <ul style="list-style-type: none"> <li>Coverage improvement for the anchor cell of a low-frequency SCG: supported only in LTE FDD</li> <li>Other functions: no difference</li> </ul> | <a href="#">4 NSA Anchor Coverage Extension Based on User Experience</a>     |

## Differences Between NR FDD and NR TDD

| Function Name                                              | Difference | Chapter/Section                                                              |
|------------------------------------------------------------|------------|------------------------------------------------------------------------------|
| NSA carrier combination selection based on user experience | None       | <a href="#">3 NSA Carrier Combination Selection Based on User Experience</a> |
| NSA anchor coverage extension based on user experience     | None       | <a href="#">4 NSA Anchor Coverage Extension Based on User Experience</a>     |

## Differences Between NSA and SA

| Function Name                                              | Difference                       | Chapter/Section                                                              |
|------------------------------------------------------------|----------------------------------|------------------------------------------------------------------------------|
| NSA carrier combination selection based on user experience | Supported only in NSA networking | <a href="#">3 NSA Carrier Combination Selection Based on User Experience</a> |
| NSA anchor coverage extension based on user experience     | Supported only in NSA networking | <a href="#">4 NSA Anchor Coverage Extension Based on User Experience</a>     |

## Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

| Function Name                                              | Difference                                  | Chapter/Section                                                              |
|------------------------------------------------------------|---------------------------------------------|------------------------------------------------------------------------------|
| NSA carrier combination selection based on user experience | Supported only in low frequency bands of NR | <a href="#">3 NSA Carrier Combination Selection Based on User Experience</a> |

| Function Name                                          | Difference | Chapter/Section                                                          |
|--------------------------------------------------------|------------|--------------------------------------------------------------------------|
| NSA anchor coverage extension based on user experience | None       | <a href="#">4 NSA Anchor Coverage Extension Based on User Experience</a> |

# 3 NSA Carrier Combination Selection Based on User Experience

---

## 3.1 Principles

### Basic Principles

After an SCG is added for an NSA UE in NSA networking, the number of carriers on the LTE side of the UE may be less than the number of carriers in LTE CA. In this case, the UE may have worse downlink experience in EN-DC than in LTE CA. **NSA carrier combination selection based on user experience can be used to provide the best service experience for the NSA UE. The eNodeB can select the most suitable combination from EN-DC or LTE CA combinations supported by the UE based on the EN-DC capability and LTE CA capability reported by the UE as well as the signal strength, load, available bandwidth, and other factors of LTE and NR cells.**

Currently, only a few NSA UEs on a live network have worse downlink experience in EN-DC than in LTE CA. Therefore, it is recommended that a blacklist or whitelist of this function be configured for NSA UEs on the eNodeB. Otherwise, negative impacts are likely to occur because all NSA UEs on the network will perform a large number of inter-frequency measurements.

This function takes effect when all of the following options are selected:

- **NSA\_LTE\_SEL\_OPT\_SW** option of the **gNodeBParam.NetworkingOptionOptSw** parameter on the NR side
- **NSA\_LTE\_FDD\_SEL\_OPT\_SW** or **NSA\_LTE\_TDD\_SEL\_OPT\_SW** option of the **EnodebAlgoExtSwitch.MultiNetworkingOptionOptSw** parameter on the LTE side

This function supports the blacklist and whitelist mechanisms. It is recommended that a blacklist or whitelist be configured on the eNodeB according to the following principles:

- If this function takes effect only for a few NSA UEs of a certain type on the network, it is recommended that the **NSA\_LTE\_SEL\_OPT\_SW\_ON** option of



the **UeCompat.WhiteLstCtrlSwitch** parameter be selected to configure a whitelist for this type of NSA UEs.

- If this function does not take effect only for a few NSA UEs of a certain type on the network, it is recommended that the **NSA\_LTE\_SEL\_OPT\_SW** option of the **UeCompat.BlacklistControlExtSwitch1** parameter be selected to configure a blacklist for this type of NSA UEs.

If both a blacklist and a whitelist are configured for a type of NSA UEs on the eNodeB, the whitelist mechanism takes effect.

The process of NSA carrier combination selection based on user experience is divided into two phases:

1. The eNodeB obtains the available bandwidths of NR cells. For details, see [Obtaining the Available Bandwidths of NR Cells](#).
2. The eNodeB performs NSA carrier combination selection based on user experience. For details, see [NSA Carrier Combination Selection Based on User Experience](#).

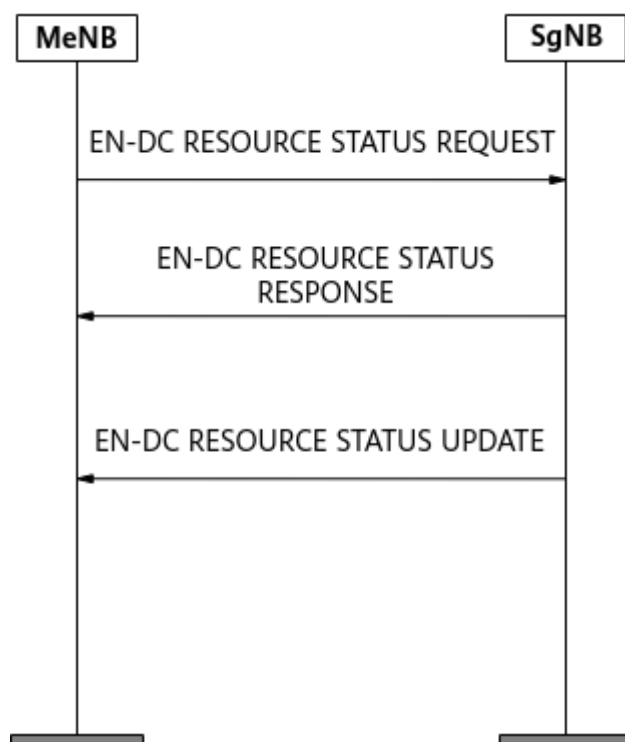
#### NOTE

If there are multiple carrier combinations in each band combination, the eNodeB can select a carrier combination from multiple carrier combinations in a band combination, in addition to selecting a band combination from multiple band combinations. This document describes how the eNodeB selects a carrier combination based on user experience.

## Obtaining the Available Bandwidths of NR Cells

**Figure 3-1** shows the procedure for the eNodeB to obtain the available bandwidths of NR cells.

**Figure 3-1** Procedure for obtaining the available bandwidths of NR cells



The procedure is described as follows:

1. The MeNB sends an EN-DC RESOURCE STATUS REQUEST message to the SgNB, requesting the SgNB to estimate the available bandwidths and loads of NR cells.
2. The SgNB sends an EN-DC RESOURCE STATUS RESPONSE message to the MeNB, starts the evaluation, and sends the evaluation result to the MeNB through an EN-DC RESOURCE STATUS UPDATE message.
3. Subsequently, the SgNB sends the latest available bandwidths of NR cells to the MeNB through an EN-DC RESOURCE STATUS UPDATE message every 10s.

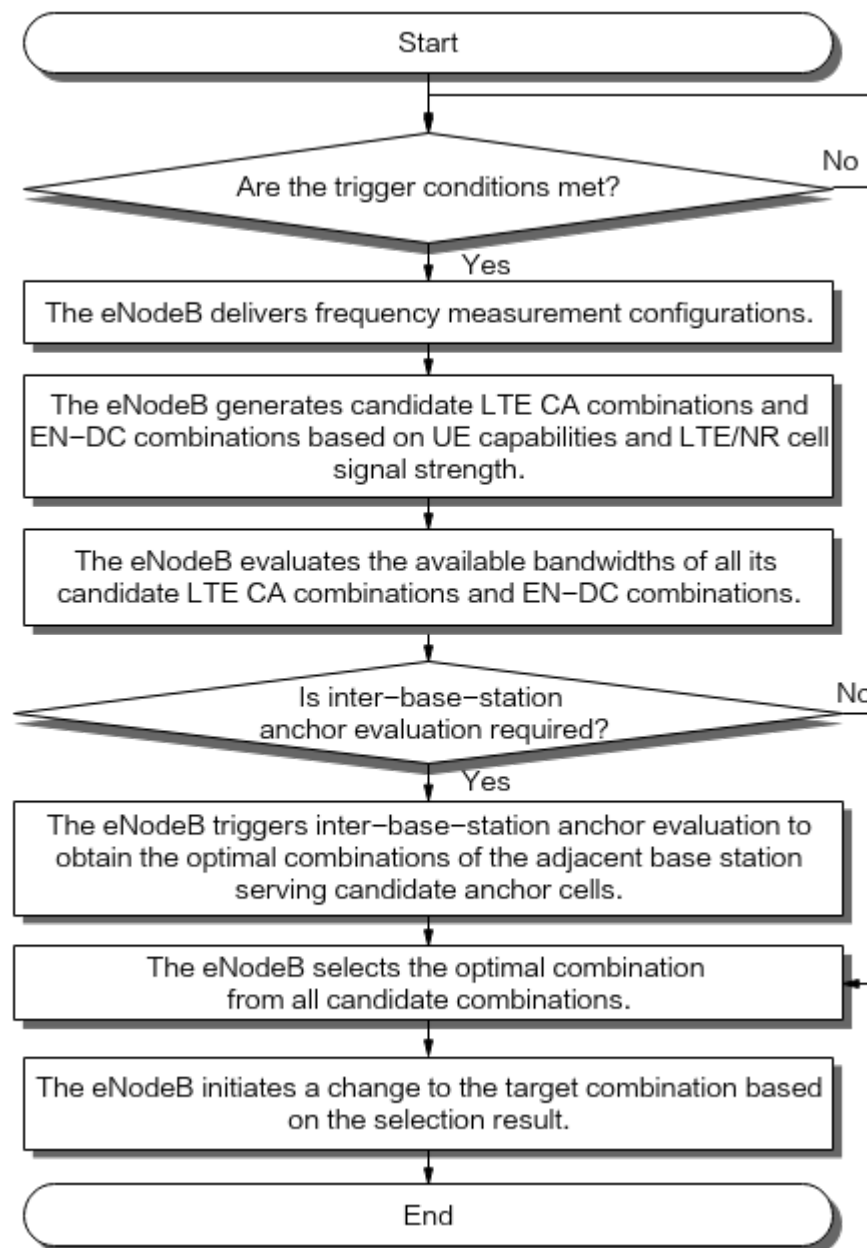
If NR CA does not take effect, the MeNB can evaluate the single-NR-carrier capability of an EN-DC combination based on the information (such as the available bandwidths and loads of NR cells) obtained through this procedure.

#### NOTE

If the spectral efficiency of the NR network is higher than that of the LTE network, the NR network can deliver better experience than the LTE network for the same bandwidth. In this case, the available bandwidth of an NR cell estimated by the SgNB can be scaled up by a certain proportion. If the spectral efficiency of the NR network is lower than that of the LTE network, the available bandwidth of an NR cell estimated by the SgNB can be scaled down by a certain proportion. The scale-up or scale-down proportion is specified by the **NRCellMobilityConfig.NrCellDLExperienceCoeff** parameter.

## NSA Carrier Combination Selection Based on User Experience

**Figure 3-2** shows the procedure for performing NSA carrier combination selection based on user experience.

**Figure 3-2** Procedure for performing NSA carrier combination selection based on user experience

The procedure is described as follows:

1. The eNodeB checks whether the NSA UE meets the conditions for triggering NSA carrier combination selection based on user experience.

In the following scenarios, the eNodeB triggers NSA carrier combination selection based on user experience, and the procedure directly goes to 2. In other scenarios, the eNodeB continues to check whether the NSA UE meets the trigger conditions.

- The NSA UE meets the trigger conditions of SCG addition described in *EPC-based NSA Basics* or the trigger conditions of NSA PCC anchoring described in *EPC-based NSA Performance Enhancement*.

- The band combination change procedure is triggered by an NR inter-frequency handover. For the trigger conditions, see *NSA Mobility Management*.
- The band combination change procedure is triggered during coverage-based NR SCC release when NSA carrier combination selection triggered by coverage-based NR SCC release is in effect.

NSA carrier combination selection triggered by coverage-based NR SCC release takes effect when all of the following conditions are met:

- The **NR\_SCC\_REL\_NSA\_CARR\_SEL\_SW** option of the **NRCCellCaMgmtConfig.CaStrategySwitch** parameter is selected on the NR side.
- The **NR\_SCC\_REL\_NSA\_CARR\_SEL\_SW** option of the **NsaDcMgmtConfig.NsaDcAlgoEnhSwitch** parameter is selected on the LTE side.
- The frequency numbers specified by **NrScgFreqConfig.ScgDLArfcn** for which the **NR\_SCC\_REL\_NSA\_CARR\_SEL\_FLAG** option of the **NrScgFreqConfig.AggregationAttribute** parameter on the LTE side is selected contain the frequency number of an NR SCC that is released based on coverage.

After NSA carrier combination selection triggered by coverage-based NR SCC release takes effect, the carrier selection penalty timer upon NR SCC release (specified by the **NsaDcMgmtConfig.NrScRelCarrSelPunishTimer** parameter) is started on the LTE side. Before this timer expires, the band combination change procedure during coverage-based NR SCC release will not be triggered again for the NSA UE.

#### NOTE

When the NSA UE meets the conditions for triggering an SCC addition procedure, smart carrier selection based on virtual grids is triggered. For details, see *Multi-carrier Unified Scheduling* in eRAN feature documentation.

2. The eNodeB generates a measurement frequency set based on the EN-DC capability and LTE CA capability reported by the NSA UE as well as the PCC, SCG, and SCC frequency information configured on the LTE side. Then, the eNodeB obtains the information about cells on each frequency based on LTE/NR virtual grids or UE measurement results. NSA PCC and related SCG frequencies are specified by the **NrScgFreqConfig.PccDLEarfcn** and **NrScgFreqConfig.ScgDLArfcn** parameters, respectively. LTE CA PCC and related SCC frequencies are configured as described in *Carrier Aggregation* in eRAN feature documentation.
  - Based on LTE/LNR virtual grids
 

When the **NSA\_LTE\_SEL\_OPT\_WITH\_VG\_SW** option of the **LnrCarrierSelection.LnrCarrierSelectionSw** parameter is selected, the eNodeB can query LTE/LNR virtual grids to obtain the RSRP values and spectral efficiency of cells on each LTE/NR frequency. This reduces the impact of gap-assisted measurement on UE performance and shortens the duration of NSA carrier combination selection. Note that LNR virtual grids can be used for building RSRP prediction models, but not spectral efficiency prediction models. [Table 3-1](#) lists the switches to be turned on for building and using virtual grid models.

If the RSRP values of cells on LTE/NR frequencies cannot be obtained based on virtual grids or the values predicted based on virtual grids are not reliable, the RSRP values of these cells are obtained from UE measurement results.

**Table 3-1** Switches required for building and using virtual grid models

| Building and Use of Virtual Grid Models                                                                      | Switch to Be Turned On                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Building of LTE RSRP prediction models                                                                       | <p>LTE RSRP prediction models can be built when any of the following options is selected:</p> <ul style="list-style-type: none"> <li>• <b>SMART_CARRIER_SELECTION_SW</b>, <b>LOW_FREQ_SCS_OPT_SW</b>, or <b>LTE_RSRP_VG_MDL_BUILD_SW</b> option of the <b>MultiCarrUnifiedSch.MultiCarrierUnifiedSchSw</b> parameter</li> <li>• <b>NSA_ANCHOR_VIRTUAL_GRID_SW<sup>a</sup></b> option of the <b>LnrCarrierSelection.LnrCarrierSelectionSw</b> parameter</li> </ul>  |
| Building of LTE spectral efficiency prediction models                                                        | <p>When building of LTE RSRP prediction models is enabled, spectral efficiency prediction models can be built if either of the following options is selected:</p> <ul style="list-style-type: none"> <li>• <b>LTE_SPEC_EFF_VG_MDL_BUILD_SW</b> option of the <b>MultiCarrUnifiedSch.MultiCarrierUnifiedSchSw</b> parameter</li> <li>• <b>SMART_CARRIER_SELECTION_ENH_SW</b> option of the <b>MultiCarrUnifiedSch.MultiCarrierUnifiedSchSw</b> parameter</li> </ul> |
| Building of LNR RSRP prediction models                                                                       | <b>LNR_VIRTUAL_GRID_SW</b> option of the <b>LnrCarrierSelection.LnrCarrierSelectionSw</b> parameter                                                                                                                                                                                                                                                                                                                                                                |
| Use of LTE RSRP prediction models, LTE spectral efficiency prediction models, and LNR RSRP prediction models | <b>NSA_LTE_SEL_OPT_WITH_VG_SW</b> option of the <b>LnrCarrierSelection.LnrCarrierSelectionSw</b> parameter                                                                                                                                                                                                                                                                                                                                                         |

| Building and Use of Virtual Grid Models                                                                                                                                                                                        | Switch to Be Turned On |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| a: After the <b>NSA_ANCHOR_VIRTUAL_GRID_SW</b> option of the <b>LnrCarrierSelection.LnrCarrierSelectionSw</b> parameter is selected, RSRP prediction models are built only for the frequency corresponding to the NSA LTE PCC. |                        |

#### NOTE

When downlink CA is enabled on the NR side, it is recommended that the **NsaDcAlgoParam.NsaDcNrCaBwEvalInd** parameter be set to **NR\_EVAL\_ALLOWED** on the LTE side. If an NR frequency serves as a downlink SCC-only frequency for NR CA, an **NrScgFreqConfig** MO for this frequency must be added, and the **NR\_DL\_SCC\_FREQ\_ONLY\_IND** option of the **NrNFreq.AggregationAttribute** parameter for this frequency must be selected. The reason is that the eNodeB needs to obtain the RSRP value of this downlink SCC-only frequency to evaluate the EN-DC combinations containing this frequency. Since neighbor relationships are not required for cells on this downlink SCC-only frequency during NSA carrier combination selection, you are advised to select the **VG\_MODEL\_FORBID\_BUILD\_FLAG** option of the **NrNFreq.AggregationAttribute** parameter to exclude this frequency for virtual grid building.

If a UE reports an SCGFailureInformationNR message with the cause value "randomAccessProblem" or "synchReconfigFailureSCG" and the corresponding NR cell is predicted as a good candidate by virtual grid models, there remains a high probability that the cell will again fail to be added to the SCG, resulting in an increased SCG service drop rate. To mitigate this, you are advised to enable the function "penalty on virtual-grid-based addition for SCG addition failures when NSA/LTE selection is enabled". Once the function takes effect, the NR cell associated with the UE-reported SCGFailure message will be excluded from virtual-grid-based SCG addition within the penalty period. The function "penalty on virtual-grid-based addition for SCG addition failures when NSA/LTE selection is enabled" is controlled by the **NSA\_SEL\_SCG\_FAIL\_VG\_PUNISH\_SW** option of the **EnodebAlgoExtSwitch.NsaDcAlgoExtSwitch** parameter.

– Based on UE measurement results

The eNodeB delivers A5 and B1 measurement configurations for LTE and NR frequencies, starts a waiting timer, and awaits measurement reports from the UE. If **NsaDcAlgoParam.LnrCoMeasLteMeasDuration** is set to **0**, the timer duration equals the larger value among the **NrScgFreqConfig.NrB1ReportWaitingTimer** parameter values of all SCG frequencies and 3. Otherwise, the timer duration equals the larger value among the **NrScgFreqConfig.NrB1ReportWaitingTimer** parameter values of all SCG frequencies and the **NsaDcAlgoParam.LnrCoMeasLteMeasDuration** parameter value.

- If the eNodeB receives measurement reports on all of LTE frequencies and neighboring NR frequencies before the waiting timer expires, the procedure goes to [3](#).

- If the eNodeB receives measurement reports on some of LTE frequencies and neighboring NR frequencies before the waiting timer expires, then:

If the **NsaDcAlgoParam.LaaMeasGuaranteeTimer** parameter is set to a value other than **0** and the following conditions are met when the waiting timer expires:

- A conflict between NR and LAA has occurred during the measurement.
- There are LAA frequencies to be measured.
- The delivered B1 measurement configurations are triggered only by user-experience-based NSA carrier combination selection.

Then, the base station removes all measurement configurations except those of LAA frequencies delivered for NSA carrier combination selection based on user experience, delivers to-be-measured LAA frequencies within specifications, and starts a timer specified by the **NsaDcAlgoParam.LaaMeasGuaranteeTimer** parameter to wait for the UE to send measurement reports.

- If the eNodeB receives measurement reports on all of LAA frequencies before the timer expires, the procedure goes to **3**.
- If the eNodeB receives measurement reports on some of LAA frequencies before the timer expires, the procedure goes to **3** when the timer expires.

If the **NsaDcAlgoParam.LaaMeasGuaranteeTimer** parameter is set to **0**, the procedure goes to **3** when the waiting timer expires.

- If no measurement report on any frequency is received when the waiting timer expires, the procedure ends except for the following scenario: The "blind SCG addition for NSA UEs without B1 measurement results on NR frequencies" function is enabled, and the frequency with the highest SCG frequency priority can only be added based on measurement while a lower-priority frequency can be added blindly. In this case, the eNodeB can blindly add the NR frequency that does not have the highest priority. The "blind SCG addition for NSA UEs without B1 measurement results on NR frequencies" function is controlled by the **NSA\_NO\_NR\_B1\_BLIND\_ADD\_SCG\_SW** option of the **EnodebAlgoExtSwitch.NsaDcAlgoExtSwitch** parameter.

 NOTE

- An eNodeB can deliver measurement configurations for a maximum of 16 LTE frequencies and a maximum of 8 NR frequencies.
- If an LTE cell in a measurement report sent by the NSA UE is not configured in the neighboring LTE cell list of the current anchor cell and the **NsaLteSelOptIntraRatEventAnrSw** option of the **ENodeBAlgoSwitch.AnrSwitch** parameter is selected, then the eNodeB triggers an intra-RAT ANR procedure to add this neighboring LTE cell. If an NR cell in a measurement report sent by the NSA UE is not configured in the neighboring NR cell list of the current anchor cell and the **NSA\_LTE\_SEL\_OPT\_NR\_ANR\_SW** option of the **ENodeBAlgoExtSwitch.AnrOptSwitch** parameter is selected, then the eNodeB triggers an inter-RAT ANR procedure to add this neighboring NR cell.
- The **NsaDcAlgoParam.ScgFastAddUeType** parameter can be set to specify the type of UEs for which SCG addition is triggered immediately once the eNodeB has obtained the RSRP of the NR frequency with the highest SCG frequency priority.
- For UEs with an SCG added, if the frequencies in the B1 measurement configurations contain only NR PSCell or SCell frequencies, the "non-gap-assisted B1 measurement of NR SCG frequencies for NSA DC UEs" function can be enabled to improve UE throughputs. This function is controlled by the **NSA\_DC\_NR\_SCG\_B1\_NO\_GAP\_SW** option of the **ENodeBAlgoExtSwitch.NsaDcAlgoExtSwitch** parameter.
- When the **NSA\_DC\_NR\_SCG\_FBD\_B1\_NO\_GAP\_SW** option of the **UeCompat.BlacklistControlExtSwitch2** parameter is selected, "non-gap-assisted B1 measurement of NR SCG frequencies for NSA DC UEs" is forbidden for blacklisted UEs.

When the "B1 measurement optimization for the serving NR frequency in NSA networking" function is enabled and the eNodeB receives an A3 or A5 measurement report containing measurement results of the NR PSCell or SCell along with the strongest intra-frequency neighboring cell of the NR PSCell or SCell prior to delivering B1 measurement configurations, then the eNodeB will not deliver B1 measurement configurations for the NR PSCell or SCell frequency. This reduces the number of NR B1 measurements. The "B1 measurement optimization for the serving NR frequency in NSA networking" function is controlled by the **NSA\_NR\_SER\_FREQ\_B1\_MEAS\_OPT\_SW** option of the **ENodeBAlgoExtSwitch.MultiNetworkingOptionOptSw** parameter.

During UE measurement, the eNodeB also delivers intra-frequency measurement configurations related to events A1, A3, and A6 for the PCell and configured SCells. If all A5 and B1 measurement reports are received but intra-frequency measurement reports remain pending before the UE measurement waiting timer expires, the eNodeB continues to wait for the remaining intra-frequency reports. As a result, it cannot immediately proceed to candidate carrier combination generation. To address this, the **CellMultiCarrUniSch.StopMeasAfterInterFMrTmr** parameter, which specifies the waiting time for stopping measurement after all inter-frequency measurement reports are received, can be set. The eNodeB behavior depends on the parameter value as follows:

- If this parameter is set to **0**, the eNodeB immediately proceeds to candidate carrier combination generation upon receiving all A5 and B1 measurement reports.
- If this parameter is set to a value ranging from **1** to **60**, the eNodeB starts the intra-frequency measurement waiting timer (duration = parameter



value x 100 ms) upon receiving all A5 and B1 measurement reports. It proceeds to candidate carrier combination generation upon receiving all intra-frequency measurement reports, expiration of the intra-frequency measurement waiting timer, or expiration of the UE measurement waiting timer.

- If this parameter is set to **255**, the eNodeB proceeds to candidate carrier combination generation upon receiving all intra-frequency measurement reports or expiration of the UE measurement waiting timer.
- 3. The eNodeB generates candidate LTE CA combinations and candidate EN-DC combinations based on factors such as the signal strength of LTE and NR cells and the combinations supported by the NSA UE. Then, the procedure goes to [4](#).
- 4. The eNodeB estimates the available bandwidths of all its candidate LTE CA combinations and EN-DC combinations. If the measured candidate anchor cell is not served by the local base station, the procedure goes to [5](#) for inter-base-station anchor evaluation; otherwise, the procedure goes to [6](#).
  - The eNodeB estimates the available bandwidths of all its candidate LTE CA combinations.
  - The eNodeB estimates the total available bandwidths of all its candidate EN-DC combinations in the following scenarios:
    - Estimating the single-NR-carrier capabilities of EN-DC combinations  
The eNodeB only needs to estimate the single-NR-carrier capabilities of EN-DC combinations when either of the following conditions is met: (1) on the LTE side, the **NsaDcAlgoParam.NsaDcNrCaBwEvalInd** parameter is set to **NR\_EVAL\_NOT\_ALLOWED**; (2) on the LTE side, the **NsaDcAlgoParam.NsaDcNrCaBwEvalInd** parameter is set to **NR\_EVAL\_NOT\_ALLOWED\_IN\_SCG\_ADD**, and the estimation is performed when an SCG is to be added for the UE in the LTE-only state. Specifically, the eNodeB independently estimates the available NR bandwidths of EN-DC combinations based on the information such as the available bandwidths and loads of NR cells obtained according to [Obtaining the Available Bandwidths of NR Cells](#), and then estimates the total available bandwidths of EN-DC combinations by adding the available NR bandwidths to the available LTE bandwidths.
    - Estimating the multi-NR-carrier capabilities of EN-DC combinations  
The gNodeB assists in estimating the multi-NR-carrier capabilities of EN-DC combinations when either of the following conditions is met: (1) on the LTE side, the **NsaDcAlgoParam.NsaDcNrCaBwEvalInd** parameter is set to **NR\_EVAL\_ALLOWED**; (2) on the LTE side, the **NsaDcAlgoParam.NsaDcNrCaBwEvalInd** parameter is set to **NR\_EVAL\_NOT\_ALLOWED\_IN\_SCG\_ADD**, and the estimation is performed when an SCG has been added for the UE.
      - The eNodeB sends a proprietary X2 message to the gNodeB to request the estimation of the available NR bandwidths of EN-DC combinations. The message contains the UE capability information, EN-DC combinations to be estimated, and NR frequency measurement results.

Assume that only one frequency is configured in an NR frequency band. Generally, for the same NR frequency, the NR-side A5 threshold for configuring a cell on this frequency as an SCell is lower than the LTE-side B1 threshold for adding a cell on this frequency as an SCG cell. To make it easier to select an NSA carrier combination containing NR CA frequencies during user experience evaluation of the "NSA carrier combination selection based on downlink user experience" function and improve downlink user experience, it is recommended that the NSA NR SCC selection optimization function be enabled. When this function is enabled, a proprietary X2 message contains all measurement results of NR frequencies reported by UEs. When this function is disabled, the proprietary X2 message contains only the measurement results of NR frequencies reported by UEs that meet the NR SCG addition threshold. NSA NR SCC selection optimization is controlled by the **NSA\_NR\_SCC\_SEL\_OPT\_SW** option of the **EnodebAlgoExtSwitch.NsaDcAlgoExtSwitch** parameter.

The "optimized experience evaluation for NSA carrier combination selection based on user experience" function is an enhancement of NSA NR SCC selection optimization. After the "optimized experience evaluation for NSA carrier combination selection based on user experience" function is enabled, the eNodeB sends the NR cell measurement results reported by UEs and indicates whether the NR cells meet the B1 threshold for SCG addition. NR cells that do not meet the SCG addition threshold may also be selected as NR SCells. However, the gNodeB returns only the evaluation results of EN-DC combinations that meet the NR SCG addition threshold. The "optimized experience evaluation for NSA carrier combination selection based on user experience" function is controlled by the **NSA\_NR\_EVAL\_REQ\_NR\_MEAS\_IND\_SW** option of the **EnodebAlgoExtSwitch.NsaDcAlgoExtSwitch** parameter on the LTE side and the **EXP\_BASED\_NSA\_PCC\_SEL\_OPT\_SW** option of the **gNodeBAlgo.NrSmartMultiCarrSw** parameter on the NR side. The "optimized experience evaluation for NSA carrier combination selection based on user experience" function can be enabled independently of NSA NR SCC selection optimization.

The "experience evaluation for the neighboring candidate NR PCell with the highest RSRP in NSA networking" function is an enhancement of the "optimized experience evaluation for NSA carrier combination selection based on user experience" function. After the former function is enabled, the gNodeB returns only the evaluation result of the NR PCell with the highest RSRP. The "experience evaluation for the neighboring candidate NR PCell with the highest RSRP in NSA networking" function is controlled by the **NSA\_NR\_STRONGEST\_PCELL\_EVAL\_SW** option of the **EnodebAlgoExtSwitch.NsaDcAlgoExtSwitch**.

After the "optimized NSA NR SCC grid-based evaluation" function is enabled, if the RSRP of an NR SCell is predicted based on the virtual grid model, the measurement result of the NR

SCell will be included in the X2 proprietary message even if the NR SCell RSRP does not meet the NR PSCell good-point threshold for grid-based prediction. When this function is disabled, the X2 proprietary message does not include the measurement result of the NR SCell. The "optimized NSA NR SCC grid-based evaluation" function is controlled by the **NSA\_NR\_SCC\_GRID\_EVAL\_OPT\_SW** option of the **ENodeBAlgoExtSwitch.NsaDcAlgoExtSwitch** parameter.

- The gNodeB starts to estimate the available NR bandwidths after receiving the request message from the eNodeB, and then sends the estimation results to the eNodeB through a proprietary X2 message.

It is recommended that the **NSA\_BLIND\_CFG\_SCC\_NOT\_EVAL\_SW** option of the **NRCellCaMgmtConfig.CaStrategySwitch** parameter be selected, since blind-configurable NR SCCs may have no coverage. With this setting, the gNodeB excludes these SCCs when estimating available bandwidths based on the NR frequency measurement results contained in the proprietary X2 message. This ensures more accurate user experience evaluation.

For an NSA UE configured with an NR SCC, if the X2 proprietary message does not contain the measurement result of this NR SCC, or the **NSA\_BLIND\_CFG\_SCC\_NOT\_EVAL\_SW** option of the **NRCellCaMgmtConfig.CaStrategySwitch** is selected and this NR SCC is blindly configured for the UE, then the configured NR SCC will also participate in user experience evaluation when the function allowing NR SCCs configured for NSA UEs to participate in experience evaluation (controlled by the **NSA\_ACTUAL\_CFG\_SCC\_EVAL\_SW** option of the **gNodeBParam.NetworkingOptionOptSw** parameter) is enabled.

The available bandwidths are estimated based on the following factors:

- Equivalent downlink cell bandwidth: This bandwidth is calculated based on factors such as the downlink cell bandwidth, UE-supported downlink bandwidths, slot configuration, and whether the SCC supports SRS carrier switching.
- NR spectral efficiency: The NR spectral efficiency is approximately calculated based on the downlink experience coefficient of NR cells, which is specified by the **NRCellMobilityConfig.NrCellDLExperienceCoeff** parameter.
- Downlink load factor: This factor is calculated based on the number of UEs to be scheduled in the downlink per second. The downlink load factor is considered during user experience evaluation on the LTE side only when the **NsaDcMgmtConfig.NsaMinDLAvgToBeSchedUeNum** parameter is set to a non-zero value.
- Inter-gNodeB CA: When the **NSA\_SA\_SEL\_INTER\_GNB\_CA\_SW** option of the **gNodeBParam.NetworkingOptionOptSw** parameter is selected, the gNodeB can estimate the multi-carrier capabilities in inter-gNodeB CA scenarios. In this scenario, the

**NRCellCaMgmtConfig.InterGnbExpEvalCoeff** parameter can be configured to reduce the impact of inter-gNodeB delay on available bandwidth estimation.

#### NOTE

The eNodeB can estimate the available bandwidths of a maximum of 384 EN-DC combinations.

When estimating the available bandwidth of an LTE CA combination or the available LTE bandwidth in an EN-DC combination, the eNodeB can scale up or down the estimated available bandwidth based on the value of the

**CellMLB.CellCapacityScaleFactor** parameter. The value **10** is recommended for the LTE network when user experience is preferentially considered; that is, the eNodeB performs the estimation based on the actual LTE cell bandwidth. If other values are used, user experience may deteriorate.

For UEs that do not support TM9, it is recommended that the MBSFN subframe bandwidth evaluation optimization function be enabled to accurately evaluate the downlink equivalent bandwidth in LTE MBSFN subframes. This function is controlled by the **CARRIER\_SEL\_MBSFN\_BW\_EVAL\_SW** option of the **EnodebAlgoExtSwitch.NsaDcAlgoExtSwitch** parameter.

The **RSVDSWPARAM4\_BIT18** option of the **gNBRsvd.RsvdSwParam4** parameter can be used to specify whether to perform experience evaluation only for the cell with the highest RSRP among the candidate PCells and SCells that meet the conditions for experience evaluation for NSA carrier combination selection based on user experience. If this option is selected, experience evaluation is performed only for the cell with the highest RSRP. If this option is deselected, experience evaluation is performed for all the preceding cells.

5. The eNodeB triggers inter-base-station anchor evaluation to obtain the optimal combinations of the adjacent base station serving candidate anchor cells.
  - a. The eNodeB sends a proprietary X2 message to the adjacent base station, requesting to evaluate the optimal combinations. This message contains the UE capabilities and the measurement results of LTE and NR frequencies.
  - b. The adjacent base station starts NSA carrier combination selection based on user experience after receiving the request from the local eNodeB, and sends the information about the optimal combinations to the local eNodeB through a proprietary X2 message.

#### NOTE

Inter-base-station anchor evaluation requires the configuration of inter-frequency relationships between PCCs of the adjacent base station by using the **EutranInterNFreq** MO.

6. The eNodeB performs the following operations based on the setting of **NrNFreq.NrFreqAbsolutePriority** for NR frequencies in the candidate EN-DC combinations:
  - Scenario 1: If there are candidate EN-DC combinations that include NR frequencies with the **NrNFreq.NrFreqAbsolutePriority** parameter set to **ABSOLUTE\_PRIOR**, the optimal combination will be selected from these combinations.
  - Scenario 2: If there are no candidate EN-DC combinations that include NR frequencies with the **NrNFreq.NrFreqAbsolutePriority** parameter set to **ABSOLUTE\_PRIOR** but there are combinations that include NR frequencies with the parameter set to **PRIOR**, the optimal combination will be selected from these combinations.

- Scenario 3: If the **NrNFreq.NrFreqAbsolutePriority** parameter is set to **NORMAL** for the NR frequencies in all candidate EN-DC combinations, the optimal combination will be selected from these combinations and LTE CA combinations.

The eNodeB uses the following principles for selection:

- The optimal EN-DC combination is selected from all candidate EN-DC combinations based on certain factors, mainly available bandwidth, NSA PCC anchoring priority, PCell handover, and SCG frequency priority, in descending order of weight. In scenarios 1 and 2, the procedure goes to 7 after the optimal EN-DC combination is selected.

#### NOTE

- Available bandwidth: The available bandwidth of an EN-DC combination is considered to be greater than that of another combination only if the former is  $N (= 1 + \text{NsaDcAlgoParam.NsaDcBwDiffThld})$  times the latter. Otherwise, it is considered that there is no difference in the available bandwidth, and the optimal combination needs to be further determined based on other factors.
  - NSA PCC anchoring priority: It is specified by the **PccFreqCfg.NsaPccAnchoringPriority** parameter.
  - PCell handover: Combinations that are not involved in PCell handover have higher priorities.
  - SCG frequency priority: It is specified by the **NrScgFreqConfig.ScgDLArfcnPriority** parameter.
  - The threshold of the bandwidth difference between EN-DC combinations for fast-moving UEs is specified by the **NsaDcAlgoParam.FastMovUeNsaDcBwDiffThld** parameter. For more information about fast-moving UEs, see *Multi-carrier Unified Scheduling* in eRAN feature documentation.
- The optimal LTE CA combination is selected from all candidate LTE CA combinations. For details, see *Multi-carrier Unified Scheduling* in eRAN feature documentation.
  - The final optimal combination is selected from the optimal EN-DC combination and optimal LTE CA combination based on the following principles, and then the procedure goes to 7.
    - If the combination currently used for the UE is the optimal LTE CA combination, then it remains unchanged when the available bandwidth of the optimal LTE CA combination is greater than that of the optimal EN-DC combination multiplied by  $(1 + \text{NsaDcAlgoParam.NsaLteBwDiffThld} - \text{NsaDcAlgoParam.NsaLteStableBwDiffThld})$ . In other cases, the optimal EN-DC combination is selected.
    - If the combination currently used for the UE is the optimal EN-DC combination, then it changes to the optimal LTE CA combination when the available bandwidth of the optimal LTE CA combination is greater than that of the optimal EN-DC combination multiplied by  $(1 + \text{NsaDcAlgoParam.NsaLteBwDiffThld} + \text{NsaDcAlgoParam.NsaLteStableBwDiffThld})$ . In other cases, the currently used optimal EN-DC combination remains unchanged.
    - If the combination currently used for the UE is neither the optimal LTE CA combination nor the optimal EN-DC combination, then the optimal LTE CA combination is selected when its available bandwidth

is greater than that of the optimal EN-DC combination multiplied by  $(1 + \text{NsaDcAlgoParam.NsaLteBwDiffThld})$ , or the optimal EN-DC combination is selected in other cases.

 NOTE

The threshold of the bandwidth difference between an EN-DC combination and an LTE CA combination for fast-moving UEs is specified by the **NsaDcAlgoParam.FastMovUeNsaLteBwDiffThld** parameter. For more information about fast-moving UEs, see *Multi-carrier Unified Scheduling* in eRAN feature documentation.

7. The eNodeB initiates a change towards the selected combination.
- If the signal levels of the target cell for a handover and its intra-frequency neighboring cell are similar, the handover may fail. If NSA PCC anchoring for NSA carrier selection optimization based on user experience is triggered for NSA UEs and the **NsaDcAlgoParam.NsaGFactorThldForHoTrgCell** parameter is set to a value other than **32767**, an LTE inter-frequency neighboring cell can be selected as the target cell for the handover only when it meets the conditions specified by the parameter, thereby reducing the handover failure probability.

- If the EN-DC combination is selected, then:

▪ If no SCG is added for the UE, the eNodeB initiates an SgNB addition procedure towards the gNodeB corresponding to the selected EN-DC combination. In this case, there is a possibility that an intra- or inter-eNodeB handover occurs on the LTE side. For details about MeNB-initiated SgNB addition, see *NSA Mobility Management*.

▪ If an SCG has been added for the UE, the eNodeB initiates a change towards the selected EN-DC combination based on whether the LTE PCC or NR PSCC needs to be changed, as listed in [Table 3-2](#). For details, see *NSA Mobility Management*.

Table 3-2 Combination change for a UE with an SCG

| Scenario      | No NR Change                                                                   | Intra-gNodeB Intra-Frequency Handover | Intra-gNodeB Inter-Frequency Handover | Inter-gNodeB Handover                                                  |
|---------------|--------------------------------------------------------------------------------|---------------------------------------|---------------------------------------|------------------------------------------------------------------------|
| No LTE Change | See MeNB-initiated SgNB modification if there is a carrier combination change. | See MeNB-initiated SgNB modification. | See MeNB-initiated SgNB modification. | See MeNB-initiated SgNB addition and MeNB/SgNB-initiated SgNB release. |

| Scenario              | No NR Change                                                   | Intra-gNodeB Intra-Frequency Handover                          | Intra-gNodeB Inter-Frequency Handover                          | Inter-gNodeB Handover                                       |
|-----------------------|----------------------------------------------------------------|----------------------------------------------------------------|----------------------------------------------------------------|-------------------------------------------------------------|
| Intra-eNodeB Handover | See MeNB-initiated intra-MeNB handover without an SgNB change. | See MeNB-initiated intra-MeNB handover without an SgNB change. | See MeNB-initiated intra-MeNB handover without an SgNB change. | See MeNB-initiated intra-MeNB handover with an SgNB change. |
| Inter-eNodeB Handover | See MeNB-initiated inter-MeNB handover without an SgNB change. | See MeNB-initiated inter-MeNB handover without an SgNB change. | See MeNB-initiated inter-MeNB handover without an SgNB change. | See MeNB-initiated inter-MeNB handover with an SgNB change. |

#### NOTE

- MeNB-initiated intra-SgNB intra-frequency handover requires the selection of the **MENB\_TRIG\_INTRA\_SGNB\_CHANGE\_SW** option of the **NRCellNsaDcConfig.NsaDcAlgoSwitch** parameter on the NR side.
  - MeNB-initiated intra-SgNB inter-frequency handover requires the selection of both the **MENB\_TRIG\_INTRA\_SGNB\_CHANGE\_SW** option of the **NRCellNsaDcConfig.NsaDcAlgoSwitch** parameter and the **SIMU\_INTRASGNB\_INTERFREQ\_HO\_SW** option of the **gNodeBParam.NsaDcOptSwitch** parameter on the NR side.
  - When the eNodeB determines to trigger an NR PSCell change procedure with the LTE PCell unchanged and detects that the UE is performing an NR inter-frequency measurement (for example, the measurement during virtual grid model building within the NR system), the setting of the **NSA\_LTE\_SEL\_NR\_GAP\_BC\_CHG\_SW** option of the **EnodebAlgoExtSwitch.NsaDcAlgoExtSwitch** parameter determines whether the carrier combination change can take place. If this option is selected, the NR PSCell change procedure with the LTE PCell unchanged can take effect during NR gap-assisted measurement. If this option is deselected, this procedure cannot take effect during NR gap-assisted measurement.
- If the LTE CA combination is selected, then:
- If no SCG has been added for the UE, no SCG-related procedure is involved. For details about the LTE CA combination change procedure, see *Multi-carrier Unified Scheduling* in eRAN feature documentation.
  - If an SCG has been added for the UE, the eNodeB initiates an SCG removal procedure. For details, see *NSA Mobility Management*. For details about the LTE CA combination change procedure, see *Multi-carrier Unified Scheduling* in eRAN feature documentation.



 NOTE

- Assume that this function obtains the signal strength of LTE and NR frequencies based on UE measurements. Then, if the **PERIODIC\_PCC\_ANCHORING\_SW** option of the **NsaDcMgmtConfig.NsaDcAlgoSwitch** parameter is selected or SCG addition is periodically triggered after this function selects an optimal LTE CA combination, you are advised to set the **NsaDcMgmtConfig.ScgAdditionInterval** parameter based on actual scenarios, considering the impact of measurement gaps on UE performance. A larger value of this parameter results in a smaller impact of measurement gaps on UE performance.
- The MeNB can use only RSRP measurement results when selecting a target NR cell in the current version. Therefore, the **NRCellMobilityConfig.MeasTrigQuantity** parameter must be set to **RSRP** in NSA networking. Otherwise, if a UE immediately reports RSRQ-based event A2 on the NR side after an EN-DC combination change, the SgNB will immediately trigger a coverage-based inter-frequency handover.
- If an NSA UE does not support the full bandwidth of an NR cell but supports the downlink minimum carrier bandwidth (**NRDUCellBwp.DlMinCarrierBw**) configured for the NR cell, the eNodeB can estimate the total available bandwidths of EN-DC combinations for the NSA UE based on this specified downlink minimum carrier bandwidth.
- It is recommended that the RSRP threshold for event B1 related to NR SCG addition on the LTE side be higher than the RSRP threshold for event A2 related to coverage-based intra-NR inter-frequency handovers on the NR side. The purpose is to prevent the immediate triggering of a coverage-based intra-NR inter-frequency switching of the PSCell when the PSCell has changed due to an EN-DC combination selection, leading to ping-pong switching and deteriorated user experience.

## 3.2 Network Analysis

### 3.2.1 Benefits

This function enables the eNodeB to select combinations with larger downlink bandwidths for NSA UEs. This increases the downlink rates of NSA UEs and the average downlink PRB utilization of the LTE network.

### 3.2.2 Impacts

The impacts between this function and other functions are described in [3.3.2.3 Function Impacts](#).

- If SCG addition or NSA PCC anchoring for UEs in connected mode has been enabled before this function is enabled, the combination selected by this function may be different from that selected previously. As a result, the distribution and status of NSA UEs on different bands will change, resulting in changes in certain indicators on both LTE and NR sides. Such indicators include NSA handover success rate, NSA service drop rate, random access success rate, CCE aggregation level, and uplink and downlink throughput.
- This function considers the available downlink bandwidth during carrier combination selection. After this function is enabled, uplink user experience is likely to deteriorate.
- After this function takes effect:
  - The delay from initial access of NSA UEs to SgNB addition completion is prolonged. That is, the delay from the time when the MeNB receives an



RRConnectionRequest message from the UE to the time when the MeNB sends an SgNB Reconfiguration Complete message to the SgNB is prolonged. The specific delay depends on the number of measured frequencies, X2 delay, and message processing delay of the peer base station.

- The SgNB addition delay for NSA UEs in a cell increases. Specifically, the values of **L.NsaDc.SgNBAdd.ScgSplit.TimeAvg** and **L.NsaDc.SgNBAdd.ScgSplit.TimeMax** increase. This is because after receiving an SgNB Addition Request Acknowledge message from the SgNB, the eNodeB needs to calculate and generate the configuration information of the LTE SCC in the EN-DC combinations based on the CA configurations on the eNodeB side.
- This function increases the values of **L.NsaDc.MeasCtrl.NR** and **L.Signal.Num.Uu** in certain scenarios:
  - In scenarios where NSA DC is enabled, the eNodeB periodically triggers NSA carrier combination selection after selecting an optimal LTE CA combination. Before NSA carrier combination selection based on user experience takes effect, the eNodeB directly initiates SCG addition in areas with NR coverage and does not perform the periodic selection procedure.
  - In scenarios where NSA PCC anchoring is enabled, NSA carrier combination selection based on user experience still triggers NSA PCC anchoring even when there is no NSA PCC with a higher priority. Before NSA carrier combination selection based on user experience takes effect, there is a possibility that the eNodeB does not trigger the NSA PCC anchoring procedure.

As the eNodeB needs to wait for all frequency measurement reports, the value of **L.NsaDc.Meas.NR.Timeout** increases.

- This function may cause NSA-capable UEs to select LTE CA when there is NR coverage. As a result, the number of NSA-capable UEs with no SCG added increases on the LTE network, and the number of NSA UEs decreases on the NR network. This imposes the following impacts on performance counters.

LTE counters:

- The number of UEs in the NSA DC state decreases. Specifically, the values of **L.Traffic.User.NsaDc.PCell.Avg**, **L.Traffic.User.NsaDc.PCell.Max**, **L.Traffic.User.NsaDc.PCell.Avg.PLMN**, and **L.Traffic.User.NsaDc.PCell.Max.PLMN** decrease.
- The number of SCG split bearers decreases. Specifically, the values of **L.NsaDc.ScgSplitBearer.Avg** and **L.NsaDc.ScgSplitBearer.Max** decrease.
- The number of NSA UEs with no SCG added increases. Specifically, the value of **L.NsaDc.Capable.5gUser.NoScg.BorderUE.Avg** increases.
- The number of PRBs used by NSA UEs on the LTE side increases. Specifically, the values of **L.ChMeas.PRB.UL.NsaDc.Used.Avg**, **L.ChMeas.PRB.DL.NsaDc.Used.Avg**,

**L.ChMeas.PRB.UL.NsaDc.Used.Avg.PLMN**, and  
**L.ChMeas.PRB.DL.NsaDc.Used.Avg.PLMN** increase.

- For NSA UEs, the total downlink data volume transmitted over MCG bearers increases, and the total downlink and uplink data volumes transmitted over data split bearers decrease.

NR counters:

- The number of UEs in the NSA DC state decreases. Specifically, the values of **N.User.NsaDc.PSCell.Avg** and **N.User.NsaDc.PSCell.Max** decrease.
  - The number of SN-terminated MCG bearer UEs decreases. Specifically, the values of **N.User.gNodeB.SNTrmt.MCGBearer.Avg** and **N.User.gNodeB.SNTrmt.MCGBearer.Max** decrease.
  - The number of QCI-specific active UEs decreases in NSA networking. Specifically, the values of **N.User.RRCCConn.Active.QCI.Avg** and **N.User.RRCCConn.Active.QCI.Max** decrease.
- This function may reduce the number of UEs that enter the NSA DC state. As a result, there are possible decreases in the values of the following counters related to SgNB-initiated SgNB modification, change, or release:

LTE counters:

- Counters related to SgNB-initiated SgNB modification, for example, **L.NsaDc.SCG.Mod.Required.Att**
- Counters related to SgNB-initiated SgNB change, for example, **L.NsaDc.SCG.Change.Att**
- Counters related to SgNB-initiated SgNB release, for example, **L.NsaDc.SgnbTrig.SgNB.NormRel**

NR counters:

- Counters related to SgNB-initiated SgNB modification, for example, **N.NsaDc.SgNB.Mod.Required.Att**
  - Counters related to SgNB-initiated inter- or intra-SgNB PSCell change, for example, **N.NsaDc.IntraSgNB.PSCell.Change.Att** and **N.NsaDc.InterSgNB.PSCell.Change.Att**
  - Counters related to SgNB-initiated SgNB release, for example, **N.NsaDc.SgNB.Rel.SgNBTrigger**
  - Counters related to SgNB-initiated DRB release, for example, **N.NsaDc.DRB.Rel.SgNBTrigger.Coverage**
- After this function takes effect, frequency-priority-based or service-based inter-frequency handover on the NR side does not take effect for NSA UEs. Therefore, the numbers of PSCell changes for such handover will decrease. The related counters include **N.NsaDc.InterFreq.FreqPri.PSCell.Change.Att** and **N.NsaDc.InterFreq.Service.PSCell.Change.Att**.
- After this function is triggered by the eNodeB, it is more likely to conflict with the intra-RAT handover procedure triggered by the gNodeB,

resulting in a possible decrease in the PSCell change success rate.

Therefore, it is recommended that the

**LTE\_HO\_DISCARD\_NR\_REPORT\_SW** option of the **EnodebAlgoExtSwitch.NsaDcAlgoSwitch** parameter on the LTE side be selected to reduce procedure conflicts.

- After this function is triggered by the eNodeB, the PSCell modification success rate and PSCell change success rate may decrease due to conflicts between the LTE RRC connection reestablishment procedure and the gNodeB-initiated SCG modification or SCG change procedure. To reduce such procedure conflicts, enabling the function "optimized handling of procedure conflicts upon RRC connection reestablishments of NSA UEs" is recommended. This function is controlled by the **NSA\_REEST\_FLOW\_CONFLICT\_OPT\_SW** option of the **EnodebAlgoExtSwitch.NsaDcAlgoExtSwitch** parameter. For details about this function, see *EPC-based NSA Basics*.
- The CPU load of the LTE and NR main control boards increases, and the user-plane CPU usage of the LTE BBPs increases.
- The throughput of LTE-only UEs decreases when all resources on the network are occupied.
- This function may decrease the number of UEs that enter the NSA DC state and increase the number of EN-DC combination changes for UEs in the NSA DC state. As a result, the values of the **L.NsaDc.SCG.Mod.Req.Att**, **L.NsaDc.SCG.Mod.Req.Succ**, and **L.Signal.Num.X2** counters decrease or increase.
- The number of gap-assisted measurements increases, which causes possible increases in the downlink packet delay (calculated by dividing **L.Traffic.DL.PktDelay.Time** by **L.Traffic.DL.PktDelay.Num**). In addition, the average downlink UE throughput on the LTE side will decrease.
- If the **NSA\_SA\_SEL\_INTER\_GNB\_CA\_SW** option of the **gNodeBParam.NetworkingOptionOptSw** parameter is selected, the number of NR SCG addition attempts increases in NSA networking.
- After non-gap-assisted B1 measurement of NR SCG frequencies for NSA DC UEs is enabled, these UEs perform B1 measurement on NR SCG frequencies without relying on measurement gaps, therefore experiencing higher throughputs.
- After the "B1 measurement optimization for the serving NR frequency in NSA networking" function is enabled, the number of times B1 measurement configurations are delivered and the number of times B1 measurements time out decrease, and the uplink and downlink throughputs of NSA UEs increase.
- After the "blind SCG addition for NSA UEs without B1 measurement results on NR frequencies" function is enabled, the number of NSA SCG addition attempts increases, the NSA SCG addition success rate decreases, and the proportion of NR CA UEs in NSA networking increases.
- After the "experience evaluation for the neighboring candidate NR PCell with the highest RSRP in NSA networking" function is enabled, the EN-DC band combination with the NR cell having the highest RSRP can be selected for NSA DC UEs. This improves downlink user experience and increases the NR handover and SCG addition success rates.

- After the **NsaDcAlgoParam.NsaGFactorThldForHoTrgCell** parameter is set to a value other than **32767**, the number of NSA PCC anchoring times, the proportion of LTE RRC connection reestablishments, the number of times NSA carrier selection based on user experience takes effect, and the throughput gains brought by NSA carrier selection based on user experience decrease.
- After the "optimized NSA NR SCC grid-based evaluation" function is enabled, the proportion of NR CA UEs increases.
- After the function "penalty on virtual-grid-based addition for SCG addition failures when NSA/LTE selection is enabled" is enabled, the number of virtual-grid-based SCG addition failures and the number of SCG additions decrease, while the success rate of virtual-grid-based SCG additions increases.
- After the function allowing NR SCCs configured for NSA UEs to participate in experience evaluation is enabled, the average downlink throughput of NSA UEs increases.
- If the waiting time for stopping measurement after all inter-frequency measurement reports are received is set too short, the eNodeB may initiate candidate carrier combination generation before intra-frequency measurement reports are fully received. Although NSA carrier combination selection based on user experience is accelerated, the resulting combination may be suboptimal due to incomplete intra-frequency measurement reports, potentially leading to reduced UE throughput.

## 3.3 Requirements

### 3.3.1 Licenses

| RAT     | Feature ID  | Feature Name                                                         | Model        | Sales Unit |
|---------|-------------|----------------------------------------------------------------------|--------------|------------|
| LTE FDD | MRFD-171223 | NSA Carrier Combination Selection Based on User Experience (LTE FDD) | LT1SNSALTE00 | per eNodeB |
| LTE TDD | MRFD-171233 | NSA Carrier Combination Selection Based on User Experience (LTE TDD) | LT4SCSEHSTD  | per eNodeB |

| RAT | Feature ID  | Feature Name                                                    | Model        | Sales Unit |
|-----|-------------|-----------------------------------------------------------------|--------------|------------|
| NR  | MRFD-171263 | NSA Carrier Combination Selection Based on User Experience (NR) | NR0S0LSBUE00 | per gNodeB |

#### NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists (FDD)*, *License Control Item Lists (CIoT)*, and *License Control Item Lists (TDD)* in eRAN feature documentation, as well as *License Control Item Lists* in 5G RAN feature documentation.

## 3.3.2 Software

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

### 3.3.2.1 Prerequisite Functions

| RAT                | Function Name     | Function Switch                                                                                   | Reference                                                | Description |
|--------------------|-------------------|---------------------------------------------------------------------------------------------------|----------------------------------------------------------|-------------|
| LTE FDD<br>LTE TDD | Adaptive CA       | <b>FreqCfgSwitch</b><br>option of the <b>ENodeBAlgoSwitch.CaAlgoSwitch</b> parameter              | <i>Carrier Aggregation</i> in eRAN feature documentation | None        |
| LTE FDD<br>LTE TDD | NSA DC capability | <b>NSA_DC_CAPABILITY_SWITCH</b><br>option of the <b>NsaDcMgmtConfig.NsaDcAlgoSwitch</b> parameter | <i>EPC-based NSA Basics</i>                              | None        |
| LTE FDD<br>LTE TDD | NSA PCC anchoring | <b>NSA_PCC_ANCHORING_SWITCH</b><br>option of the <b>NsaDcMgmtConfig.NsaDcAlgoSwitch</b> parameter | <i>EPC-based NSA Performance Enhancement</i>             | None        |

| RAT                                          | Function Name                                              | Function Switch                                                                                                                              | Reference                                        | Description                                                                                                                                                                                                                                                                                 |
|----------------------------------------------|------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NR<br>FDD<br>Low -<br>frequency<br>NR<br>TDD | NSA carrier combination selection based on user experience | <b>NSA_LTE_SEL_OPT_SW</b> option of the <b>gNodeBParam.NetworkingOptionOptSw</b> parameter                                                   | <i>EPC-based NSA Experience Improvement</i>      | If the <b>NSA_SA_SEL_INTER_GNB_CA_SW</b> option of the <b>gNodeBParam.NetworkingOptionOptSw</b> parameter is selected, the <b>NSA_LTE_SEL_OPT_SW</b> , <b>NSA_SA_DL_SEL_OPT_SW</b> , or <b>NSA_SA_UL_SEL_OPT_SW</b> option of this parameter must be selected.                              |
| NR<br>FDD<br>Low -<br>frequency<br>NR<br>TDD | NSA/SA selection based on downlink traffic volume          | <b>NSA_SA_DL_SEL_OPT_SW</b> option of the <b>gNodeBParam.NetworkingOptionOptSw</b> parameter                                                 | <i>NSA-SA Selection Based on User Experience</i> |                                                                                                                                                                                                                                                                                             |
| NR<br>FDD<br>Low -<br>frequency<br>NR<br>TDD | NSA/SA selection based on uplink coverage                  | <b>NSA_SA_UL_SEL_OPT_SW</b> option of the <b>gNodeBParam.NetworkingOptionOptSw</b> parameter                                                 | <i>NSA-SA Selection Based on User Experience</i> |                                                                                                                                                                                                                                                                                             |
| LTE<br>FDD<br>LTE<br>TDD                     | NSA carrier combination selection based on user experience | <b>NSA_LTE_FDD_SEL_OPT_SW</b> or <b>NSA_LTE_TDD_SEL_OPT_SW</b> option of the <b>EnodebAlgoExtSwitch.MultiNetworkingOptionOptSw</b> parameter | <i>EPC-based NSA Experience Improvement</i>      | If the <b>NR_SCC_REL_NS_A_CARR_SEL_SW</b> option of the <b>NsaDcMgmtConfig.NsaDcAlgoEnhSwitch</b> parameter is selected, the <b>NSA_LTE_FDD_SEL_OPT_SW</b> or <b>NSA_LTE_TDD_SEL_OPT_SW</b> option of the <b>EnodebAlgoExtSwitch.MultiNetworkingOptionOptSw</b> parameter must be selected. |

| RAT                                                                      | Function Name                                                                                  | Function Switch                                                                                                                                                                                                                                      | Reference                                   | Description                                                                                                                                                                                                                                |
|--------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NR<br>FDD<br>Low -<br>frequency<br>NR<br>TDD                             | NSA carrier combination selection based on user experience                                     | <b>NSA_LTE_SEL_OPT_SW</b> option of the <b>gNodeBParam.NetworkingOptionOptSw</b> parameter                                                                                                                                                           | <i>EPC-based NSA Experience Improvement</i> | If the <b>NR_SCC_REL_NS_A_CARR_SEL_SW</b> option of the <b>NRCellCaMgmtConfig.CaStrategySwitch</b> parameter is selected, the <b>NSA_LTE_SEL_OPT_SW</b> option of the <b>gNodeBParam.NetworkingOptionOptSw</b> parameter must be selected. |
| LTE<br>FDD<br>LTE<br>TDD<br>NR<br>FDD<br>Low -<br>frequency<br>NR<br>TDD | Optimized experience evaluation for NSA carrier combination selection based on user experience | <b>NSA_NR_EVAL_REQ_NR_MEAS_IN_D_SW</b> option of the <b>EnodebAlgoExtSwitch.NsaDcAlgoExtSwitch</b> parameter on the LTE side and the <b>EXP_BASED_NSA_PCC_SEL_OPT_SW</b> option of the <b>gNodeBAlgo.NrSmartMultiCarrSw</b> parameter on the NR side | <i>EPC-based NSA Experience Improvement</i> | To enable "experience evaluation for the neighboring candidate NR PCell with the highest RSRP in NSA networking", "optimized experience evaluation for NSA carrier combination selection based on user experience" must be enabled.        |
| LTE<br>FDD<br>LTE<br>TDD<br>NR<br>FDD<br>Low -<br>frequency<br>NR<br>TDD | Optimized experience evaluation for NSA carrier combination selection based on user experience | <b>NSA_NR_EVAL_REQ_NR_MEAS_IN_D_SW</b> option of the <b>EnodebAlgoExtSwitch.NsaDcAlgoExtSwitch</b> parameter on the LTE side and the <b>EXP_BASED_NSA_PCC_SEL_OPT_SW</b> option of the <b>gNodeBAlgo.NrSmartMultiCarrSw</b> parameter on the NR side | <i>EPC-based NSA Experience Improvement</i> | To enable "optimized NSA NR SCC grid-based evaluation", "optimized experience evaluation for NSA carrier combination selection based on user experience" must be enabled.                                                                  |

3.3.2.2 Mutually Exclusive Functions

None

3.3.2.3 Function Impacts

| RAT                | Function Name                                         | Function Switch                                                                                                                                                                                     | Reference                                                      | Description                                                                                                                                                                                                                                                                                                                                             |
|--------------------|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LTE FDD<br>LTE TDD | Spectrum coordination                                 | <b>SpectrumCoordinationSwitch</b> option of the <b>CaMgtCfg.CellCaAlgoSwitch</b> parameter, and <b>NSA_DC_FLEXIBLE_PCC_ANCHOR_SW</b> option of the <b>NsaDcMgmtConfig.NsaDcAlgoSwitch</b> parameter | <i>LTE Spectrum Coordination</i> in eRAN feature documentation | This function does not take effect for NSA UEs in a cell where NSA carrier combination selection based on user experience has taken effect.                                                                                                                                                                                                             |
| LTE FDD<br>LTE TDD | Exclusion of load factors during bandwidth evaluation | <b>BW_EVAL_EXCLUDE_LOAD_SW</b> option of the <b>eNodeBUSParaCfg.NsaDcUsUeCarrSelAlgoSw</b> parameter                                                                                                | None                                                           | After exclusion of load factors during bandwidth evaluation takes effect, the eNodeB performs NSA carrier combination selection based on user experience only for NSA DC UEs running unlimited services and the eNodeB does not consider cell load when evaluating the available bandwidths of LTE CA combinations and EN-DC combinations for such UEs. |



| RAT                      | Function Name                                                            | Function Switch                                                                                                                             | Reference                                                             | Description                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------|--------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LTE<br>FDD<br>LTE<br>TDD | Threshold adaptation for inter-frequency handovers based on LTE coverage | <b>FDD_ANCHOR_ADAPT_COV_THLD_SW</b> or <b>TDD_ANCHOR_ADAPT_COV_THLD_SW</b> option of the <b>NsaDcAlgoParam.NsaDcAlgoExtSwitch</b> parameter | <i>EPC-based NSA Experience Improvement</i>                           | After receiving an event A2 measurement report related to coverage-based inter-frequency handovers from a target UE for coverage extension, the eNodeB immediately triggers NSA carrier combination selection based on user experience. The LTE CA band combination of the target UE for coverage extension may change.                                      |
| LTE<br>FDD<br>LTE<br>TDD | Optimization-based carrier selection                                     | <b>OPT_MODEL_BASED_CARR_SEL_SW</b> option of the <b>MultiCarrUnifiedSch.MultiCarrierUnifiedSchEnSw</b> parameter                            | <i>Multi-carrier Unified Scheduling</i> in eRAN feature documentation | When the NSA PCC anchoring function is enabled and the NSA carrier combination selection based on user experience function is disabled in NSA networking, optimization-based carrier selection manages only SCC selection, rather than PCC anchoring, for NSA UEs. As such, the gains of optimization-based carrier selection may decrease in this scenario. |
| LTE<br>FDD<br>LTE<br>TDD | NSA PCC anchoring                                                        | <b>NSA_PCC_ANCHORING_SWITCH</b> option of the <b>NsaDcMgmtConfig.NsaDcAlgoSwitch</b> parameter                                              | <i>EPC-based NSA Performance Enhancement</i>                          |                                                                                                                                                                                                                                                                                                                                                              |

| RAT                      | Function Name                                                                     | Function Switch                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Reference                                           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LTE<br>FDD<br>LTE<br>TDD | PCI confusion detection based on the SCG addition success rate                    | <ul style="list-style-type: none"> <li>LTE:<br/><b>NR_EVENT_ANR_SW</b> option of the <b>CellAlgoSwitch.AnrFunctionSwitch</b> parameter<br/><b>NCellParaCfg.CellAnrOptMode</b> set to <b>FREE</b>.<br/><b>NSA_SCG_ADD_SCR_NCELL_MGMT_SW</b> option of the <b>ENodeBAlgoExtSwitch.AnrOptSwitch</b> parameter<br/><b>ANR.NoHoSetMode</b> set to <b>PCI_CHECK</b></li> <li>NR:<br/><b>NSA_SCG_ADD_SCR_NCELL_MGMT_SW</b> option of the <b>gNodeBAlgo.SelfConfigSwitch</b> parameter</li> </ul> | <i>ANR Management</i> in eRAN feature documentation | <p>When NSA carrier combination selection based on user experience is enabled, UEs cannot read the cell global identifiers (CGIs) of neighboring cells and therefore, PCI confusion detection based on the SCG addition success rate does not take effect.</p> <p>If PCI confusion detection based on the SCG addition success rate needs to take effect in this case, select the <b>NSA_LTE_SEL_OPT_NR_ANR_SW</b> option of the <b>ENodeBAlgoExtSwitch.AnrOptSwitch</b> parameter.</p> |
| LTE<br>FDD<br>LTE<br>TDD | FDD SCG addition for NR CA-capable UEs based on the RSRP of candidate NR TDD SCCs | <b>NsaDcAlgoParam.NrTddSccRsrpThld</b> set to a value other than <b>255</b>                                                                                                                                                                                                                                                                                                                                                                                                               | <i>EPC-based NSA Basics</i>                         | The "FDD SCG addition for NR CA-capable UEs based on the RSRP of candidate NR TDD SCCs" function does not take effect when NSA carrier combination selection based on user experience takes effect.                                                                                                                                                                                                                                                                                     |

| RAT                                     | Function Name                                   | Function Switch                                                                                                                                                                     | Reference                                                                        | Description                                                                                                                                                                                                                            |
|-----------------------------------------|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NR<br>FDD<br>Low-frequency<br>NR<br>TDD | Spectral-efficiency-based connected mode MLB    | <b>INTER_FREQ_CONNECTED_MLB_SW</b> option of the <b>NRCellAlgoSwitch</b> . <b>MlbAlgoSwitch</b> parameter                                                                           | <i>Mobility Load Balancing in Connected Mode</i> in 5G RAN feature documentation | After NSA carrier combination selection based on user experience takes effect in a cell, neither spectral-efficiency-based connected mode MLB nor UE-number-based MLB in frequency rotation mode takes effect for NSA UEs in the cell. |
| NR<br>FDD<br>Low-frequency<br>NR<br>TDD | UE-number-based MLB in frequency rotation mode  | <b>FR1_TO_FR2_FREQ_RR</b> option of the <b>NRCellAlgoSwitch</b> . <b>InterFreqHoSwitch</b> parameter                                                                                |                                                                                  |                                                                                                                                                                                                                                        |
| NR<br>FDD<br>Low-frequency<br>NR<br>TDD | UE-number-based connected mode MLB              | <b>INTER_FREQ_CONNECTED_MLB_SW</b> option of the <b>NRCellAlgoSwitch</b> . <b>MlbAlgoSwitch</b> parameter, and <b>NRCellMlb.MlbTriggerMode</b> set to <b>CELL_USER_NUM</b>          | <i>Mobility Load Balancing in Connected Mode</i> in 5G RAN feature documentation | After NSA carrier combination selection based on user experience takes effect, UE-number-based connected mode MLB does not take effect for NSA UEs in the cell.                                                                        |
| NR<br>FDD<br>Low-frequency<br>NR<br>TDD | 5QI-specific UE-number-based connected mode MLB | <b>INTER_FREQ_CONNECTED_MLB_SW</b> option of the <b>NRCellAlgoSwitch</b> . <b>MlbAlgoSwitch</b> parameter, and <b>NRCellMlb.MlbTriggerMode</b> set to <b>SPECIFIED_5QI_USER_NUM</b> | <i>Mobility Load Balancing in Connected Mode</i> in 5G RAN feature documentation | After NSA carrier combination selection based on user experience takes effect, 5QI-specific UE-number-based connected mode MLB does not take effect for NSA UEs in the cell.                                                           |
| NR<br>FDD<br>Low-frequency<br>NR<br>TDD | Experience-based smart carrier selection        | <b>MULTI_FREQ_SMART_SEL_SW</b> option of the <b>NRCellAlgoSwitch</b> . <b>MultiFreqAlgoSwitch</b> parameter                                                                         | <i>Multi-Frequency Smart Selection</i> in 5G RAN feature documentation           | When both experience-based smart carrier selection and NSA carrier combination selection based on user experience are enabled, experience-based smart carrier selection does not take effect for NSA UEs.                              |

| RAT                                     | Function Name                                             | Function Switch                                                                                                                               | Reference                                                              | Description                                                                                                                                                                                     |
|-----------------------------------------|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NR<br>FDD<br>Low-frequency<br>NR<br>TDD | Multi-frequency beam coordination                         | <b>MULTI_FREQ_BEAM_COORD_SW</b> or <b>MULTI_FREQ_BEAM_COORD_LITE_SW</b> option of the <b>NRDUCellFeaturesWithMultiCarrierSwitch</b> parameter | <i>Multi-Frequency Smart Selection</i> in 5G RAN feature documentation | When both multi-frequency beam coordination and NSA carrier combination selection based on user experience are enabled, multi-frequency beam coordination does not take effect for NSA UEs.     |
| NR<br>FDD<br>Low-frequency<br>NR<br>TDD | Uplink multi-frequency coordination                       | <b>MULTI_FREQ_UL_COORD_SW</b> option of the <b>NRDUCellFeaturesWithMultiCarrierSwitch</b> parameter                                           | <i>Multi-Frequency Smart Selection</i> in 5G RAN feature documentation | When both uplink multi-frequency coordination and NSA carrier combination selection based on user experience are enabled, uplink multi-frequency coordination does not take effect for NSA UEs. |
| NR<br>FDD<br>Low-frequency<br>NR<br>TDD | Latency-based optimal carrier selection                   | <b>DELAY_BASED_CARRIER_SEL_SW</b> option of the <b>NRCCellAlgoSwitchWithMultiFreqAlgoSwitch</b> parameter                                     | <i>Multi-Frequency Smart Selection</i> in 5G RAN feature documentation | Latency-based optimal carrier selection does not take effect for NSA UEs when it is enabled together with NSA carrier combination selection based on user experience.                           |
| NR<br>FDD<br>Low-frequency<br>NR<br>TDD | Frequency-priority-based inter-frequency handover         | <b>FREQ_PRIORITY_BASED_HO</b> option of the <b>NRCCellAlgoSwitchWithInterFreqHoSwitch</b> parameter                                           | <i>NSA Mobility Management</i>                                         | These functions do not take effect for NSA UEs in a cell where NSA carrier combination selection based on user experience has taken effect.                                                     |
|                                         | Operator-specific-priority-based inter-frequency handover | <b>OP_DED_PRIORITY_BASED_HO</b> option of the <b>NRCCellAlgoSwitchWithInterFreqHoSwitch</b> parameter                                         |                                                                        |                                                                                                                                                                                                 |

| RAT                | Function Name                                   | Function Switch                                                                                                                                                                                                                                                                       | Reference                                      | Description                                                                                                                                                                         |
|--------------------|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    | Service-based inter-frequency handover          | <b>SERVICE_BASED_HO</b> option of the <b>NRCCellAlgoSwitch</b> . <b>InterFreqHoSwitch</b> parameter                                                                                                                                                                                   |                                                |                                                                                                                                                                                     |
| LTE FDD            | DSS and NR Flexible Refarming                   | <ul style="list-style-type: none"> <li>LTE: <b>DSS_FLEX_REFARM_SW</b> option of the <b>SpectrumCloud</b>. <b>SpectrumCloudEnhSwitch</b> parameter</li> <li>NR: <b>DSS_FLEX_REFARM_SW</b> option of the <b>NRDUCCellAlgoSwitch</b>. <b>SpectrumCloudEnhSwitch</b> parameter</li> </ul> | <i>LTE FDD and NR Dynamic Spectrum Sharing</i> | After refarming from spectrum sharing to NR, the number of RBs available for NR may change and the optimal carrier combination may also change.                                     |
| LTE FDD<br>LTE TDD | SPID-based SCG frequency priority configuration | <b>SPIDCFG.NrScgFreqPriorityGroupId</b> set to a value other than <b>65535</b>                                                                                                                                                                                                        | <i>EPC-based NSA Basics</i>                    | After SPID-based SCG frequency priority configuration is enabled, NSA carrier combination selection based on user experience does not take effect for UEs with the specified SPIDs. |

| RAT                                     | Function Name       | Function Switch                                                                       | Reference                                                                          | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------------------|---------------------|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NR<br>FDD<br>Low-frequency<br>NR<br>TDD | QoS-based CA policy | <b>QOS_CA_POLICY_SW</b> option of the <b>gNodeBParam.NetworkOptionOptSw</b> parameter | <i>Service-differentiated Experience Guarantee</i> in 5G RAN feature documentation | <p>When both NSA carrier combination selection based on user experience and QoS-based CA policy are enabled, if the LTE-side parameter <b>NsaDcAlgoParam.NsaDcNrCaBwEvalInd</b> is set to <b>NR_EVAL_ALLOWED</b> or <b>NR_EVAL_NOT_ALLOWED_IN_SCG_ADD</b>, CA may not take effect for UEs with a specific QCI and using the limited frequency group because the LTE side does not transfer QCIs to the NR side. In this case, the <b>NSA_L2N_QCI_TRANSFER_SW</b> option of the <b>EnodebAlgoExtSwitch.MultiNetworkOptionOptSw</b> parameter can be selected on the LTE side. The LTE side will transfer the UE QCI list to the NR side when requesting experience evaluation assistance from the NR side, thereby avoiding affecting the QoS-based CA policy function.</p> <p>For details about MML-based configurations of the <b>NSA_L2N_QCI_TRANSFER_SW</b> option of the</p> |

| RAT | Function Name | Function Switch | Reference | Description                                                                                                                                              |
|-----|---------------|-----------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |               |                 |           | <b>EnodebAlgoExtSwitch.MultiNetworkingOptionOptSw</b> parameter, see <i>Service-differentiated Experience Guarantee</i> in 5G RAN feature documentation. |

### 3.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

#### Base Station Models

On the LTE side, the compatible base stations are as follows:

- 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R. 5900 series base stations must be configured with a BBU model other than the BBU5900C.
- DBS3900 LampSite and DBS5900 LampSite

On the NR side, the compatible base stations are as follows:

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910, and 5900 series base stations must be configured with a BBU model other than the BBU5900C.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

#### Boards

LTE:

- Main control boards: UMPTb and later
- Baseband processing units: LBBPd and later

NR: no requirements

#### RF Modules

This function does not depend on RF modules.

### 3.3.4 Networking

This function is not recommended in the following networking scenarios:

- LTE and NR are deployed in the same band. In this scenario, the carrier combination selection result may not be optimal.
- LTE and NR belong to different PLMNs. The eNodeB selects NR cells based on SCG downlink frequency priorities, available RBs, and RSRP. The PLMN of the selected NR cells is used as the target PLMN of the EN-DC combinations. This will prevent NR CA cells in another PLMN from being selected as the target SCG cells, even if the available aggregated bandwidth of such cells is greater than the available bandwidth of the selected NR cells. As a result, a non-optimal EN-DC combination may be selected.
- RAN sharing with common carrier is enabled for LTE or NR. In this scenario, the available bandwidths evaluated by the eNodeB or gNodeB are inaccurate, affecting the function performance. For details about RAN sharing with common carrier, see *RAN Sharing* in eRAN feature documentation and *Multi-Operator Sharing* in 5G RAN feature documentation.

For other networking requirements, see *EPC-based NSA Basics*.

### 3.3.5 Cells

- This function must be enabled in both the source and target PCells. If a measured candidate PCell is not served by the local base station, inter-base-station anchor evaluation is required. If this function is not enabled in the target PCell, inter-base-station anchor evaluation will fail and consequently, this function will not take effect.
- Both the source and target PCells cannot be cells served by non-Huawei base stations at the coverage border. This function does not take effect for adjacent base stations provided by different vendors because such base stations cannot exchange proprietary X2 messages required by this function.

### 3.3.6 Others

- UE
  - UEs must support NSA DC specified in 3GPP Release 15.
  - UEs must have subscribed to LTE and NR services.
  - UEs must match gNodeB and eNodeB versions.
- EPC
  - The EPC must be CloudEPC to support Option 3 and Option 3x.
  - The EPC must support NSA DC. If the core network is provided by Huawei, see *WSFD-021101 5G NSA (Opt.3) Dual Connectivity Management* for details.
  - If NSA DC is enabled on an eNodeB, the connected MMEs need to support NSA DC. If a connected MME does not support NSA DC, the **MmeCapInfo.MmeNsaDcCapability** parameter for this MME must be set to **NOT\_SUPPORT**.
- X2 transmission bandwidth

If an EN-DC combination is selected by the function of NSA carrier combination selection based on user experience, the UE may fail to use all



available bandwidths on the LTE or NR side due to X2 rate limitation after the UE is switched to the EN-DC combination. To ensure that the evaluation result of NSA carrier combination selection based on user experience meets the expectation, an appropriate maximum X2 transmission rate needs to be set. If the actual X2 transmission bandwidth cannot support the maximum air interface transmission rate on the LTE or NR side, this function is not recommended. The maximum X2 transmission rate is specified by **gNodeBParam.MaxX2TransRate** or **gNodeBParam.IntraSiteMaxX2TransRate** in Option 3x, or specified by **NsaDcAlgoParam.X2TransMaxRate** in Option 3.

## 3.4 Operation and Maintenance

### 3.4.1 When to Use

- This function is recommended when operators require selection between EN-DC and LTE CA combinations based on available bandwidths and all the following conditions are met:
- The bandwidths of LTE-only CA combinations are greater than those of EN-DC combinations. The LTE-only CA capabilities include LTE CA and LTE CA+LAA capabilities.
  - In Option 3x, the downlink dynamic data split mode is used at the PDCP layer, that is, the **gNBPDcpParamGroup.DlDataPdcpsplitMode** parameter is set to **SCG\_AND\_MCG**.
  - X2 links between eNodeBs are functional.
- The larger the bandwidth difference between LTE-only CA combinations and EN-DC combinations, the greater the gains achieved by this function.

### 3.4.2 Data Configuration

#### 3.4.2.1 Data Preparation

Before activating this function, ensure that the **NRCeIlRelation**, **NRCeIlFreqRelation**, and **NRExternalNCeIl** MOs have been configured. For details about the configuration, see *SA Mobility Management in Connected Mode* in 5G RAN feature documentation.

The following table describes the parameters that must be set in the **EnodebAlgoExtSwitch** MO on the LTE side and the **gNodeBParam** MO on the NR side to configure this function.

| Parameter Name                     | Parameter ID                                                   | Setting Notes                                                                       |
|------------------------------------|----------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Multi Networking Option Opt Switch | LTE:<br><b>EnodebAlgoExtSwitch.Multi NetworkingOptionOptSw</b> | Select the <b>NSA_LTE_FDD_SEL_OPT_SW</b> and <b>NSA_LTE_TDD_SEL_OPT_SW</b> options. |

| Parameter Name           | Parameter ID                                    | Setting Notes                                |
|--------------------------|-------------------------------------------------|----------------------------------------------|
| Networking Option Opt Sw | NR:<br><b>gNodeBParam.NetworkingOptionOptSw</b> | Select the <b>NSA_LTE_SEL_OPT_SW</b> option. |

The following table describes the parameters that must be set in the **NsaDcMgmtConfig** and **NrScgFreqConfig** MOs on the LTE side to enable NSA carrier combination selection triggered by coverage-based NR SCC release and set the penalty timer.

| Parameter Name                                | Parameter ID                                      | Setting Notes                                                                                                                                                                                                                                        |
|-----------------------------------------------|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NSA DC Algorithm Switch                       | <b>NsaDcMgmtConfig.NsaDcAlgorithmEnhSwitch</b>    | To enable NSA carrier combination selection triggered by coverage-based NR SCC release, select the <b>NR_SCC_REL_NSA_CARR_SEL_SW</b> option.                                                                                                         |
| NR SCC Release Carrier Selection Punish Timer | <b>NsaDcMgmtConfig.NrScgRelCarrSelPunishTimer</b> | Set this parameter based on the network plan.                                                                                                                                                                                                        |
| Aggregation Attribute                         | <b>NrScgFreqConfig.AggregationAttribute</b>       | To enable NSA carrier combination selection triggered by coverage-based NR SCC release for the NR SCC whose frequency number is specified by the <b>NrScgFreqConfig.ScgDlArfcn</b> parameter, select the <b>NR_SCC_REL_NSA_CARR_SEL_FLAG</b> option. |

The following table describes the parameter that must be set in the **NRCellCaMgmtConfig** MO on the NR side to enable NSA carrier combination selection triggered by coverage-based NR SCC release.

| Parameter Name     | Parameter ID                                     | Setting Notes                                                                                                                                |
|--------------------|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| CA Strategy Switch | <b>NRCellCaMgmtCon-<br/>fig.CaStrategySwitch</b> | To enable NSA carrier combination selection triggered by coverage-based NR SCC release, select the <b>NR_SCC_REL_NSA_CARR_SEL_SW</b> option. |

The following table describes the parameters that must be set in the **NsaDcAlgoParam** MO on the LTE side to configure NR CA bandwidth estimation.

| Parameter Name                              | Parameter ID                                  | Setting Notes                                                                                                                                                                                                                         |
|---------------------------------------------|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NSA DC NR CA Bandwidth Evaluation Indicator | <b>NsaDcAlgoParam.NsaDcNrCa<br/>BwEvalInd</b> | <ul style="list-style-type: none"><li>• The value <b>NR_EVAL_ALLOWED</b> is recommended if CA is enabled on the NR side.</li><li>• The value <b>NR_EVAL_NOT_ALLOWED</b> is recommended if CA is not enabled on the NR side.</li></ul> |

The following table describes the parameters that must be set in the **NrNFreq** MO on the LTE side to configure the absolute priority of an NR frequency.

| Parameter Name                 | Parameter ID                          | Setting Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------------------------|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NR Frequency Absolute Priority | <b>NrNFreq.NrFreqAbsolutePriority</b> | <ul style="list-style-type: none"><li>It is recommended that the <b>NrNFreq.NrFreqAbsolutePriority</b> parameter be set to <b>ABSOLUTE_PRIOR</b> for frequencies with high priorities specified by <b>NrScgFreqConfig.ScgDLArfcnPriority</b>.</li><li>The value <b>ABSOLUTE_PRIOR</b> is recommended for an NR frequency whose bandwidth is much greater than the LTE bandwidth.</li><li>The value <b>PRIOR</b> is recommended for an NR frequency whose bandwidth is obviously greater than the LTE bandwidth.</li></ul> |

The following table describes the parameters that must be set in the **NrNFreq** MO on the LTE side to configure a downlink NR SCC-only frequency.

| Parameter Name        | Parameter ID                        | Setting Notes                                                                                      |
|-----------------------|-------------------------------------|----------------------------------------------------------------------------------------------------|
| Aggregation Attribute | <b>NrNFreq.AggregationAttribute</b> | Select or deselect the <b>NR_DL_SCC_FREQ_ONLY_IND</b> option based on the operator's network plan. |

The following table describes the parameters that must be set in the **NsaDcAlgoParam** MO on the LTE side to configure the bandwidth difference threshold for NSA carrier combination selection based on user experience.

| Parameter Name                             | Parameter ID                                    | Setting Notes                                                                                                                                                                                 |
|--------------------------------------------|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NSA LTE Bandwidth Diff Thld                | <b>NsaDcAlgoParam.NsaLteBwDiffThld</b>          | The default value is recommended. If this parameter is set to an excessively small value, an LTE CA combination without obvious advantage will be selected, which may bring negative impacts. |
| NSA DC Bandwidth Diff Thld                 | <b>NsaDcAlgoParam.NsaDcBwDiffThld</b>           | The default value is recommended.                                                                                                                                                             |
| NSA LTE Stable Bandwidth Diff Thld         | <b>NsaDcAlgoParam.NsaLteStableBwDiffThld</b>    | Set this parameter based on the network plan.                                                                                                                                                 |
| Fast Moving UE NSA LTE Bandwidth Diff Thld | <b>NsaDcAlgoParam.FastMovUeNsaLteBwDiffThld</b> | The default value is recommended.                                                                                                                                                             |
| Fast Moving UE NSA DC Bandwidth Diff Thld  | <b>NsaDcAlgoParam.FastMovUeNsaDcBwDiffThld</b>  | The default value is recommended.                                                                                                                                                             |

The following table describes the parameter that must be set in the **NsaDcAlgoParam** MO on the LTE side to configure the LAA measurement guarantee timer.

| Parameter Name                  | Parameter ID                                | Setting Notes                                 |
|---------------------------------|---------------------------------------------|-----------------------------------------------|
| LAA Measurement Guarantee Timer | <b>NsaDcAlgoParam.LaaMeasGuaranteeTimer</b> | Set this parameter based on the network plan. |

The following table describes the parameters that must be set in the **NrCellNsaDcConfig** MO on the NR side to configure MeNB-initiated intra-SgNB intra-frequency handover.

| Parameter Name          | Parameter ID                                  | Setting Notes                                            |
|-------------------------|-----------------------------------------------|----------------------------------------------------------|
| NSA DC Algorithm Switch | <b>NrCellNsaDcConfig.NsaDcAlgorithmSwitch</b> | Select the <b>MENB_TRIG_INTRA_SGNB_CHANGE_SW</b> option. |

The following table describes the parameters that must be set in the **gNodeBParam** MO on the NR side to configure MeNB-initiated intra-SgNB inter-frequency handover.

| Parameter Name             | Parameter ID                      | Setting Notes                                            |
|----------------------------|-----------------------------------|----------------------------------------------------------|
| NSA DC Optimization Switch | <b>gNodeBParam.NsaDcOptSwitch</b> | Select the <b>SIMU_INTRASGNB_INTERFREQ_HO_SW</b> option. |

The following table describes the parameters that must be set in the **UeCompat** and **eNBRsvdPara** MOs on the LTE side as well as the **gNBRsvdExt01** MO on the NR side to configure whitelist and blacklist control for this function.

| Parameter Name                       | Parameter ID                               | Setting Notes                                                                                                                                     |
|--------------------------------------|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| White List Control Switch            | <b>UeCompat.WhiteLstCtrlSwitch</b>         | It is recommended that the <b>NSA_LTE_SEL_OPT_SW_ON</b> option be selected if this function takes effect for a small number of NSA UE types.      |
| Blacklist Control Extension Switch 1 | <b>UeCompat.BlacklistControlExtSwitch1</b> | It is recommended that the <b>NSA_LTE_SEL_OPT_SW</b> option be selected if this function does not take effect for a small number of NSA UE types. |

The following table describes the parameters that must be set in the **MultiCarrUnifiedSch** and **LnrCarrierSelection** MOs on the LTE side to configure LTE/NR virtual grid model building and application.

| Parameter Name                     | Parameter ID                                          | Setting Notes                                                                                                                                                                                                                                                                                                                                                              |
|------------------------------------|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Multi-Carrier Unified Sch Switch   | <b>MultiCarrUnifiedSch.MultiCarrierUnifiedSchSw</b>   | <ul style="list-style-type: none"> <li>To enable LTE RSRP virtual grid model building, select the <b>LTE_RSRP_VG_MDL_BUILD_SW</b> option.</li> <li>To enable LTE spectral-efficiency virtual grid model building, select the <b>LTE_SPEC_EFF_VG_MDL_BUILD_SW</b> option.</li> </ul>                                                                                        |
| LNR Carrier Selection Switch       | <b>LnrCarrierSelection.LnrCarrierSelectionSw</b>      | To enable NR RSRP virtual grid model building, select the <b>LNR_VIRTUAL_GRID_SW</b> option.                                                                                                                                                                                                                                                                               |
| LNR Carrier Selection Switch       | <b>LnrCarrierSelection.LnrCarrierSelectionSw</b>      | Select the <b>NSA_LTE_SEL_OPT_WITH_VG_SW</b> option.                                                                                                                                                                                                                                                                                                                       |
| Multi Networking Option Opt Switch | <b>EnodebAlgoExtSwitch.MultiNetworkingOptionOptSw</b> | Select the <b>NSA_NR_SER_FREQ_B1_MEAS_OPT_SW</b> option to enable B1 measurement optimization for the serving NR frequency in NSA networking, if the measurement results of NR PSCells or SCCells carried in A3 or A5 measurement reports meet conditions and the base station is expected not to deliver B1 measurement configurations for the corresponding frequencies. |
| NSA DC Algorithm Ext Switch        | <b>EnodebAlgoExtSwitch.NsaDcAlgoExtSwitch</b>         | Select the <b>NSA_SEL_SCG_FAIL_VG_PUNISH_SW</b> option to enable the function "penalty on virtual-grid-based addition for SCG addition failures when NSA/LTE selection is enabled", if the NR cell associated with the UE-reported SCGFailure message needs to be excluded from virtual-grid-based SCG addition within the penalty period.                                 |

The following table describes the parameters used for optimizing NSA carrier combination selection based on user experience.

| Parameter Name                         | Parameter ID                                        | Setting Notes                                                                                                                                                                                                  |
|----------------------------------------|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NSA DC Algorithm Switch                | <b>NsaDcAlgoParam.NsaDcAlgoSwitch</b>               | It is recommended that the <b>LAA_MEAS_GAP_PRI_SW</b> option be selected. Under this setting, NR frequencies are preferentially selected for gap-assisted measurement and LAA frequencies may not be measured. |
| NR B1 Report Waiting Timer             | <b>NrScgFreqConfig.NrB1ReportWaitingTimer</b>       | Set this parameter based on the network plan.                                                                                                                                                                  |
| LNR Co-Meas LTE Measure Duration       | <b>NsaDcAlgoParam.LnrCoMeasLteMeasDuration</b>      | Set this parameter based on the network plan.                                                                                                                                                                  |
| Stop Measure After Inter-Freq MR Timer | <b>CellMultiCarrUniSch.StopMeasAfterInterFMrTmr</b> | Set this parameter based on the network plan. If the main control board is from the UMPTb series and the BBP is from the UBBPd series, the setting of this parameter may not be effective.                     |
| Multi-Carrier Unified Sch Switch       | <b>MultiCarrUnifiedSch.MultiCarrierUnifiedSchSw</b> | Set the <b>FAST_MOVING_UE_LOAD_CALC_SW</b> option for fast-moving UEs based on the LTE network plan.                                                                                                           |
| NSA DC Algorithm Switch                | <b>NsaDcAlgoParam.NsaDcAlgoSwitch</b>               | Set the <b>FAST_MOVING_UE_NR_LOAD_CALC_SW</b> option for fast-moving UEs based on the NR network plan.                                                                                                         |
| NSA DC Algorithm Switch                | <b>NsaDcAlgoParam.ScgFastAddUeType</b>              | Set this parameter based on the actual situation.                                                                                                                                                              |
| ANR algorithm switch                   | <b>ENodeBAlgoSwitch.AnrSwitch</b>                   | It is recommended that the <b>NsaLteSelOptIntraRatEventAnrSw</b> option be selected in scenarios where intra-LTE ANR takes effect.                                                                             |



| Parameter Name                               | Parameter ID                                        | Setting Notes                                                                                                                                                                                  |
|----------------------------------------------|-----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ANR Optimization Switch                      | <b>EnodebAlgoExtSwitch.AnrOptSwitch</b>             | It is recommended that the <b>NSA_LTE_SEL_OPT_NR_ANR_SW</b> option be selected in scenarios where inter-RAT ANR with NG-RAN takes effect.                                                      |
| NR Cell Downlink Experience Coefficient      | <b>NRCeMlMobilityConfig.NrCellDlExperienceCoeff</b> | The default value is recommended.                                                                                                                                                              |
| Cell Capacity Scale Factor                   | <b>CellMLB.CellCapacityScaleFactor</b>              | The default value is recommended if user experience is preferentially considered.                                                                                                              |
| NSA Min DL Average To-Be-Scheduled UE Number | <b>NsaDcMgmtConfig.NsaMinDLAvgToBeSchedUeNum</b>    | Set this parameter if the downlink load factor needs to be considered during user experience evaluation on the LTE side. You are advised to set this parameter based on the network plan.      |
| Networking Option Opt Sw                     | <b>gNodeBParam.NetworkingOptionOptSw</b>            | Select the <b>NSA_SA_SEL_INTER_GNB_CA_SW</b> option if estimation of the multi-carrier capabilities in inter-gNodeB CA scenarios is required.                                                  |
| Inter-gNodeB Exp Evaluation Coefficient      | <b>NRCeMlCaMgmtConfig.InterGnbExpEvalCoeff</b>      | The default value is recommended.                                                                                                                                                              |
| NSA G Factor Thld for Handover Target Cell   | <b>NsaDcAlgoParam.NsaGFactorThldForHoTrgCell</b>    | Set this parameter based on the network plan. This parameter specifies the G factor threshold for target cells during handovers in NSA carrier combination selection based on user experience. |
| NSA DC Algorithm Ext Switch                  | <b>EnodebAlgoExtSwitch.NsaDcAlgoExtSwitch</b>       | Select the <b>NSA_NR_SCC_SEL_OPT_SW</b> option.                                                                                                                                                |
| NSA DC Algorithm Ext Switch                  | <b>EnodebAlgoExtSwitch.NsaDcAlgoExtSwitch</b>       | Select the <b>CARRIER_SEL_MBSFN_BW_EVAL_SW</b> option.                                                                                                                                         |

| Parameter Name                | Parameter ID                                  | Setting Notes                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NSA DC Algorithm Ext Switch   | <b>EnodebAlgoExtSwitch.NsaDcAlgoExtSwitch</b> | To enable an NR PSCell change procedure with the LTE PCell unchanged to take effect during NR gap-assisted measurement, select the <b>NSA_LTE_SEL_NR_GAP_BC_CHG_SW</b> option. With this setting, the number of NSA NR PSCell changes, the number of conflicts between NR gap-assisted measurement and random access, and the delay of UEs accessing NR cells increase. |
| NSA DC Algorithm Ext Switch   | <b>EnodebAlgoExtSwitch.NsaDcAlgoExtSwitch</b> | Select the <b>NSA_NR_EVAL_REQ_NR_MEAS_IND_SW</b> option.<br><br>The "optimized experience evaluation for NSA carrier combination selection based on user experience" function is an enhancement to the NSA NR SCC selection optimization function. Therefore, the former function can be enabled independently of the latter function.                                  |
| NR Smart Multi-Carrier Switch | <b>gNodeBAlgo.NrSmartMultiCarrierSw</b>       | Select the <b>EXP_BASED_NSA_PCC_SELECTION_SW</b> option.<br><br>The "optimized experience evaluation for NSA carrier combination selection based on user experience" function is an enhancement to the NSA NR SCC selection optimization function. Therefore, the former function can be enabled independently of the latter function.                                  |

| Parameter Name                       | Parameter ID                                  | Setting Notes                                                                                                                                                                                                                                                                 |
|--------------------------------------|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reserved Switch<br>Parameter 4       | <b>gNBRsvd.RsvdSwParam4</b>                   | To enable experience evaluation only for the cell with the highest RSRP among the candidate PCells and SCells that meet the conditions for experience evaluation for NSA carrier combination selection based on user experience, select the <b>RSVDSWPARAM4_BIT18</b> option. |
| CA Strategy Switch                   | <b>NRCeCellCaMgmtConfig.CaStrategySwitch</b>  | Select the <b>NSA_BLIND_CFG_SCC_NOT_EVAL_SW</b> option.                                                                                                                                                                                                                       |
| NSA DC Algorithm Ext Switch          | <b>EnodebAlgoExtSwitch.NsaDcAlgoExtSwitch</b> | Select the <b>NSA_DC_NR_SCG_B1_NO_GAP_SW</b> option to enable non-gap-assisted B1 measurement of NR SCG frequencies for NSA DC UEs, when the B1 measurement configuration containing only NR PSCell or SCell frequencies is not expected to contain gap configurations.       |
| Blacklist Control Extension Switch 2 | <b>UeCompat.BlacklistControlExtSwitch2</b>    | To disable "non-gap-assisted B1 measurement of NR SCG frequencies for NSA DC UEs" for blacklisted UEs, select the <b>NSA_DC_NR_SCG_FBD_B1_NO_GAP_SW</b> option.                                                                                                               |

| Parameter Name              | Parameter ID                                  | Setting Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------------|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NSA DC Algorithm Ext Switch | <b>EnodebAlgoExtSwitch.NsaDcAlgoExtSwitch</b> | In scenarios where the frequency with the highest SCG frequency priority can only be added based on measurement while a lower-priority frequency can be added blindly, if no measurement report on any frequency is received when the measurement waiting timer expires and the eNodeB is required to blindly add the NR frequency that does not have the highest priority, select the <b>NSA_NO_NR_B1_BLIND_ADD_SCG_SW</b> option to enable the "blind SCG addition for NSA UEs without B1 measurement results on NR frequencies" function. |
| NSA DC Algorithm Ext Switch | <b>EnodebAlgoExtSwitch.NsaDcAlgoExtSwitch</b> | If the gNodeB is required to return only the evaluation result of the NR PSCell with the highest RSRP during NR-assisted evaluation, select the <b>NSA_NR_STRONGEST_PCELL_EVAL_SW</b> option to enable the "experience evaluation for the neighboring candidate NR PCell with the highest RSRP in NSA networking" function.                                                                                                                                                                                                                  |
| NSA DC Algorithm Ext Switch | <b>EnodebAlgoExtSwitch.NsaDcAlgoExtSwitch</b> | If the X2 proprietary message is required to contain the measurement result of any NR SCC whose RSRP does not meet the NR PSCell good-point threshold for grid-based prediction, select the <b>NSA_NR_SCC_GRID_EVAL_OPT_SW</b> option to enable the "optimized NSA NR SCC grid-based evaluation" function.                                                                                                                                                                                                                                   |

| Parameter Name           | Parameter ID                             | Setting Notes                                                                                                                                                |
|--------------------------|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Networking Option Opt Sw | <b>gNodeBParam.NetworkingOptionOptSw</b> | To enable the function allowing NR SCCs configured for NSA UEs to participate in experience evaluation, select the <b>NSA_ACTUAL_CFG_SCC_EVAL_SW</b> option. |

### 3.4.2.2 Using MML Commands

Before using MML commands, refer to [3.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

### Activation Command Examples

On the eNodeB side

```
//Enabling NSA carrier combination selection based on user experience
MOD ENODEBALGOEXTSWITCH:
MultiNetworkingOptionOptSw=NSA_LTE_FDD_SEL_OPT_SW-1&NSA_LTE_TDD_SEL_OPT_SW-1;
//(Optional) Enabling NSA carrier combination selection triggered by coverage-based NR SCC release and
setting the penalty timer (if required)
MOD NSADCMGMTCONFIG: LocalCellId=0, NrScRelCarrSelPunishTimer=8,
NsaDcAlgoEnhSwitch=NR_SCC_REL_NSA_CARR_SEL_SW-1;
MOD NRSCGFREQCONFIG: PccDlEarfcn=1500, ScgDlArfcn=151600,
AggregationAttribute=NR_SCC_REL_NSA_CARR_SEL_FLAG-1;
//(Optional) Setting the NR CA bandwidth evaluation switch when CA is enabled on the NR side to
determine whether the MeNB is allowed to initiate an SgNB-assisted evaluation process when evaluating
the total bandwidth of NR CA cells in an NSA DC combination. In the following example, the MeNB is
allowed to initiate an SgNB-assisted evaluation process in NR CA scenarios.
MOD NSADCALGOPARAM: NsaDcNrCaBwEvalInd=NR_EVAL_ALLOWED;
//(Optional) Setting the absolute priority of an NR frequency that has a high SCG DL ARFCN priority or has
a much greater bandwidth than the LTE bandwidth
MOD NRNFREQ: LocalCellId=0, DlArfcn=151600, NrFreqAbsolutePriority=ABSOLUTE_PRIOR;
//(Optional) Setting an NR frequency as a downlink SCC-only frequency when downlink CA is enabled on
the NR side and this NR frequency needs to be used only as a downlink SCC
MOD NRNFREQ: LocalCellId=0, DlArfcn=151600, AggregationAttribute=NR_DL_SCC_FREQ_ONLY_IND-1;
//Setting the thresholds of bandwidth difference and stable-switching-oriented bandwidth difference
between LTE CA and EN-DC combinations as well as the threshold of bandwidth difference between EN-DC
combinations
MOD NSADCALGOPARAM: NsaDcBwDiffThld=10, FastMovUeNsaLteBwDiffThld=50, NsaLteBwDiffThld=50,
NsaLteStableBwDiffThld=5, FastMovUeNsaDcBwDiffThld=10;
//(Optional) Setting the LAA measurement guarantee timer based on the network plan
MOD NSADCALGOPARAM: LaaMeasGuaranteeTimer=0;
//(Optional) Setting the duration of the A5 and B1 measurement report waiting timer
MOD NRSCGFREQCONFIG: PccDlEarfcn=1500, ScgDlArfcn=151600, NrB1ReportWaitingTimer=3;
MOD NSADCALGOPARAM: LnrCoMeasLteMeasDuration=0;
//(Optional) Setting StopMeasAfterInterFMrTmr
```

```

MOD CELLMULTICARRUNISCH: LocalCellId=0, StopMeasAfterInterFMrTmr=0;
//(Optional) Setting whitelist control of this function if this function needs to take effect only for
whitelisted UEs on the network
ADD UECOMPAT: Index=0, UeInfoType=UE_CAPABILITY, UeCapIndex=0,
WhiteLstCtrlSwitch=NSA_LTE_SEL_OPT_SW_ON-1;
//(Optional) Setting blacklist control of this function if this function is not allowed to take effect for
blacklisted UEs on the network
ADD UECOMPAT: Index=0, UeInfoType=UE_CAPABILITY, UeCapIndex=0,
BlacklistControlExtSwitch1=NSA_LTE_SEL_OPT_SW-1;
//(Optional) Setting the LTE cell capacity scale factor based on the network plan. It is recommended that
this factor be set to the default value when user experience is preferentially considered.
MOD CELLMLB: LocalCellId=0, CellCapacityScaleFactor=10;
//(Optional) Setting NsaMinDlAvgToBeSchedUeNum, if the downlink load factor needs to be considered
during user experience evaluation on the LTE side
MOD NSADCMGMTCONFIG: LocalCellId=0, NsaMinDlAvgToBeSchedUeNum=255;
//(Optional) Turning on the switches for LTE/NR virtual grid model building and application if spectral
efficiency prediction models need to be built
MOD MULTICARRUNIFIEDSCH:
MultiCarrierUnifiedSchSw=LTE_RSRP_VG_MDL_BUILD_SW-1&LTE_SPEC_EFF_VG_MDL_BUILD_SW-1;
MOD LNRCARRIERSELECTION:
LnrCarrierSelectionSw=LNR_VIRTUAL_GRID_SW-1&NSA_LTE_SEL_OPT_WITH_VG_SW-1;
//(Optional) Enabling B1 measurement optimization for the serving NR frequency in NSA networking, if the
measurement results of NR PSCells or SCCells carried in A3 or A5 measurement reports meet conditions and
the base station is expected not to deliver B1 measurement configurations for the corresponding frequencies
MOD ENODEBALGOEXTSWITCH: MultiNetworkingOptionOptSw=NSA_NR_SER_FREQ_B1_MEAS_OPT_SW-1;
//(Optional) Enabling the function "penalty on virtual-grid-based addition for SCG addition failures when
NSA/LTE selection is enabled", if the NR cell associated with the NE-reported SCGFailure message needs to
be excluded from virtual-grid-based SCG addition within the penalty period
MOD ENODEBALGOEXTSWITCH: NsaDcAlgoExtSwitch=NSA_SEL_SCG_FAIL_VG_PUNISH_SW-1;
//(Optional) Setting the UE type for fast SCG addition when fast SCG addition for fast-moving UEs is
enabled
MOD NSADCALGOPARAM: ScgFastAddUeType=FAST_MOVING_UE;
//(Optional) Modifying the following parameter to set the G factor threshold for target cells during
handovers in NSA carrier combination selection based on user experience
MOD NSADCALGOPARAM: NsaGFactorThldForHoTrgCell=30;
//(Optional) Enabling NSA NR SCC selection optimization (if required)
MOD ENODEBALGOEXTSWITCH: NsaDcAlgoExtSwitch=NSA_NR_SCC_SEL_OPT_SW-1;
//(Optional) Enabling optimized experience evaluation for NSA carrier combination selection based on user
experience
MOD ENODEBALGOEXTSWITCH: NsaDcAlgoExtSwitch=NSA_NR_EVAL_REQ_NR_MEAS_IND_SW-1;
//(Optional) Enabling MBSFN subframe bandwidth evaluation optimization (if required)
MOD ENODEBALGOEXTSWITCH: NsaDcAlgoExtSwitch=CARRIER_SEL_MBSFN_BW_EVAL_SW-1;
//(Optional) Enabling an NR PSCell change procedure with the LTE PCell unchanged to take effect during
NR gap-assisted measurement
MOD ENODEBALGOEXTSWITCH: NsaDcAlgoExtSwitch=NSA_LTE_SEL_NR_GAP_BC_CHG_SW-1;
//(Optional) Selecting the NSA_DC_NR_SCG_B1_NO_GAP_SW option to enable non-gap-assisted B1
measurement of NR SCG frequencies for NSA DC UEs, when the B1 measurement configuration containing
only NR PSCell or SCCell frequencies is not expected to include gap configurations
MOD ENODEBALGOEXTSWITCH: NsaDcAlgoExtSwitch=NSA_DC_NR_SCG_B1_NO_GAP_SW-1;
//(Optional) Selecting the following option to disable "non-gap-assisted B1 measurement of NR SCG
frequencies for NSA DC UEs" for blacklisted UEs
//Running the following commands if no UeCompat MO has been configured
ADD UEINFO: UeCapIndex=0, AccessStratumRelease=1, UeCategory=40, Fgi=1247894;
ADD UECOMPAT: Index=0, UeInfoType=UE_CAPABILITY, UeCapIndex=0,
BlacklistControlExtSwitch2=NSA_DC_NR_SCG_FBD_B1_NO_GAP_SW-1;
//Running the following command if a UeCompat MO has been configured
MOD UECOMPAT: Index=0, UeInfoType=UE_CAPABILITY, UeCapIndex=0,
BlacklistControlExtSwitch2=NSA_DC_NR_SCG_FBD_B1_NO_GAP_SW-1;
//(Optional) Enabling "blind SCG addition for NSA UEs without B1 measurement results on NR
frequencies", if the eNodeB is required to blindly add an NR frequency that does not have the highest
priority
MOD ENODEBALGOEXTSWITCH: NsaDcAlgoExtSwitch=NSA_NO_NR_B1_BLIND_ADD_SCG_SW-1;
//(Optional) Enabling "experience evaluation for the neighboring candidate NR PCell with the highest RSRP
in NSA networking", if the gNodeB is required to return only the evaluation result of the NR PSCell with the
highest RSRP during NR-assisted evaluation
MOD ENODEBALGOEXTSWITCH: NsaDcAlgoExtSwitch=NSA_NR_STRONGEST_PCELL_EVAL_SW-1;
//(Optional) Enabling optimized NSA NR SCC grid-based evaluation, if the X2 proprietary message needs to
contain the measurement result of any NR SCC whose RSRP does not meet the NR PSCell good-point

```

threshold for grid-based prediction

```
MOD ENODEBALGOEXTSWITCH: NsaDcAlgoExtSwitch=NSA_NR_SCC_GRID_EVAL_OPT_SW-1;
```

### On the gNodeB side

```
//Enabling NSA carrier combination selection based on user experience
MOD GNODEBPARAM: NetworkingOptionOptSw=NSA_LTE_SEL_OPT_SW-1;
//Enabling MeNB-initiated intra-SgNB cell change
MOD NRCELLNSADCCONFIG: NrCellId=0, NsaDcAlgoSwitch=MENB_TRIG_INTRA_SGNB_CHANGE_SW-1;
MOD GNODEBPARAM: NsaDcOptSwitch=SIMU_INTRASGNB_INTERFREQ_HO_SW-1;
//Setting the downlink experience coefficient for an NR cell
MOD NRCELLMOBILITYCONFIG: NrCellId=0, NrCellDlExperienceCoeff=150;
//(Optional) Enabling NSA carrier combination selection triggered by coverage-based NR SCC release (if required)
MOD NRCELLCAMGMTCONFIG: NrCellId=0, CaStrategySwitch=NR_SCC_REL_NSA_CARR_SEL_SW-1;
//(Optional) Turning on the NSA/SA carrier selection in inter-gNodeB CA switch if estimation of multi-carrier capabilities in inter-gNodeB CA scenarios is required
MOD GNODEBPARAM: NetworkingOptionOptSw=NSA_SA_SEL_INTER_GNB_CA_SW-1;
//(Optional) Setting the coefficient used for experience evaluation in inter-gNodeB CA scenarios
MOD NRCELLCAMGMTCONFIG: NrCellId=0, InterGnbExpEvalCoeff=70;
//(Optional) Excluding blind-configurable NR SCCs from available bandwidth evaluation
MOD NRCELLCAMGMTCONFIG: NrCellId=0, CaStrategySwitch=NSA_BLIND_CFG_SCC_NOT_EVAL_SW-1;
//(Optional) Enabling optimized experience evaluation for NSA carrier combination selection based on user experience
MOD GNODEBALGO: NrSmartMultiCarrSw=EXP_BASED_NSA_PCC_SEL_OPT_SW-1;
//(Optional) Enabling experience evaluation only for the cell with the highest RSRP among the candidate PCells and SCells that meet the conditions for experience evaluation for NSA carrier combination selection based on user experience
MOD GNBRSVD: RsvdSwParam4=RSVDSWPARAM4_BIT18-1;
//(Optional) Enabling the function that allows NR SCCs configured for NSA UEs to participate in experience evaluation
MOD GNODEBPARAM: NetworkingOptionOptSw=NSA_ACTUAL_CFG_SCC_EVAL_SW-1;
```

## Optimization Command Examples

Only the eNodeB side is involved.

```
//Turning on the LAA_MEAS_GAP_PRI_SW
MOD NSADCALGOPARAM: NsaDcAlgoSwitch=LAA_MEAS_GAP_PRI_SW-1;
//Turning on the FAST_MOVING_UE_LOAD_CALC_SW
MOD MULTICARRUNIFIEDSCH: MultiCarrierUnifiedSchSw=FAST_MOVING_UE_LOAD_CALC_SW-1;
//Turning on the FAST_MOVING_UE_NR_LOAD_CALC_SW
MOD NSADCALGOPARAM: NsaDcAlgoSwitch=FAST_MOVING_UE_NR_LOAD_CALC_SW-1;
//Turning on the switches for intra-LTE event-triggered ANR and inter-RAT event-triggered ANR with NG-RAN during NSA/LTE selection
MOD ENODEBALGOSWITCH: AnrSwitch=NsaLteSelOptIntraRatEventAnrSw-1;
MOD ENODEBALGOEXTSWITCH: AnrOptSwitch=NSA_LTE_SEL_OPT_NR_ANR_SW-1;
```

## Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

### On the eNodeB side

```
//(Optional) Disabling NSA carrier combination selection triggered by coverage-based NR SCC release
MOD NRSCGFREQCONFIG: PccDlEarfcn=1500, ScgDlArfcn=151600,
AggregationAttribute=NR_SCC_REL_NSA_CARR_SEL_FLAG-0;
MOD NSADCMGMTCONFIG: LocalCellId=0, NsaDcAlgoEnhSwitch=NR_SCC_REL_NSA_CARR_SEL_SW-0;
//Disabling NSA carrier combination selection based on user experience
MOD ENODEBALGOEXTSWITCH:
MultiNetworkingOptionOptSw=NSA_LTE_FDD_SEL_OPT_SW-0&NSA_LTE_TDD_SEL_OPT_SW-0;
```

### On the gNodeB side

```
//(Optional) Turning off NSA_SA_SEL_INTER_GNB_CA_SW to disable the estimation of multi-carrier capabilities in inter-gNodeB CA scenarios
```

```
MOD GNODEBPARAM: NetworkingOptionOptSw=NSA_SA_SEL_INTER_GNB_CA_SW-0;
//(Optional) Disabling NSA carrier combination selection triggered by coverage-based NR SCC release
MOD NRCELLCAMGMTCONFIG: NrCellId=0, CaStrategySwitch=NR_SCC_REL_NSA_CARR_SEL_SW-0;
//Disabling NSA carrier combination selection based on user experience
MOD GNODEBPARAM: NetworkingOptionOptSw=NSA_LTE_SEL_OPT_SW-0;
```

### 3.4.2.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

#### NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

## 3.4.3 Activation Verification

### Using Counters

Counters related to NSA carrier combination selection based on user experience can be subscribed on the MAE-Access in one-click mode.

If any of the following counters produces a non-zero value after an NSA UE accesses the network, this function has taken effect.

| Counter ID | Counter Name              | NE     |
|------------|---------------------------|--------|
| 1526763124 | L.NsaDc.BcSelOpt.BcChange | eNodeB |
| 1526763123 | L.NsaDc.BcSelOpt.LteCa    | eNodeB |

### Using Messages

- Log in to the MAE-Access. Choose **Monitor > Signaling Trace > Signaling Trace Management**. The **Signaling Trace Management** window is displayed.
- NR cell load subscription is successful.  
Choose **Trace Type > LTE > Application Layer > X2 Interface Trace**. Check that the eNodeB sends an ENDC\_RSRC\_STATUS\_REQUEST message to the gNodeB and then the gNodeB responds with an ENDC\_RSRC\_STATUS\_RESPONSE message over the X2 interface.
- Random access is successful.  
Choose **Trace Type > LTE > Application Layer > Uu Interface Trace**. Check that the UE sends an RRC\_CONN\_SETUP\_CMP message to the eNodeB to initiate an LTE access procedure.
- (Optional) The eNodeB delivers NR measurement configurations.  
Choose **Trace Type > LTE > Application Layer > Uu Interface Trace**. Check that the eNodeB sends an RRC\_CONN\_RECFG message to the UE, carrying PCC, SCC, and SCG frequency measurement configurations.
- (Optional) The UE reports measurement results.  
Choose **Trace Type > LTE > Application Layer > Uu Interface Trace**. Check that the UE sends an RRC\_MEAS\_RPRT message, carrying the IEs related to the measured PCIs and signal strength of PCells, SCells, and NR cells.



6. (Optional) The adjacent base station determines its optimal combinations.  
Choose **Trace Type > LTE > Application Layer > X2 Interface Trace**. Check that the source eNodeB sends information such as NSA UE capabilities as well as LTE and NR cell measurement results to the adjacent eNodeB for evaluation through a proprietary X2 message, and after the evaluation, the adjacent eNodeB sends the optimal combinations to the source eNodeB through a HUAWEI\_PRIVATE\_MSG message.
7. (Optional) The eNodeB delivers a reconfiguration message indicating a change to the target combination, and the UE responds with a reconfiguration complete message.  
Choose **Trace Type > LTE > Application Layer > Uu Interface Trace**. Check that the eNodeB sends an RRC\_CONN\_RECFG message to the UE, carrying the information about the target combination.

### 3.4.4 Network Monitoring

After NSA carrier combination selection based on user experience takes effect, the following counters can be used to observe the changes in the frequency band combinations of UEs:

- **L.NsaDc.BcSelOpt.BcChange.LteCaToLteCa.DL**: measures the number of times the base station selects a different LTE CA band combination from the current one for UEs in a cell.
- **L.NsaDc.BcSelOpt.BcChange.LteCaToENDC.DL**: measures the number of times the base station selects an EN-DC band combination to replace the current LTE CA band combination for UEs in a cell.
- **L.NsaDc.BcSelOpt.BcChange.ENDCToLteCa.DL**: measures the number of times the base station selects an LTE CA band combination to replace the current EN-DC band combination for UEs in a cell.
- **L.NsaDc.BcSelOpt.BcChange.ENDCToENDC.DL**: measures the number of times the base station selects a different EN-DC band combination from the current one for UEs in a cell.

# 4 NSA Anchor Coverage Extension Based on User Experience

## 4.1 Principles

In NSA networking with an LTE medium frequency, an LTE low frequency, and an NR low frequency, when the LTE medium frequency serves as the anchor for NSA UEs, the 5G online duration of the UEs may be short and NR low frequency resources cannot be fully used due to the following reasons:

- The coverage of the LTE medium frequency is different from that of the NR low frequency.
- There is coverage for UEs in the NR low frequency. UEs are close to the coverage edge of the LTE medium frequency and therefore are handed over to the LTE low frequency through coverage-based inter-frequency handovers.
- Due to the limited UE capability, the LTE low frequency cannot serve as the NSA anchor and the NR low frequency cannot be added as an SCG frequency.

**NSA anchor coverage extension based on user experience enables the eNodeB to determine whether to allow inter-frequency handovers due to coverage or triggered by carrier selection for NSA UEs based on the principles of continuous 5G connection, good user experience, and stable network KPIs. In this way, the coverage edge of LTE medium frequencies is extended and the 5G online duration of UEs is increased.**

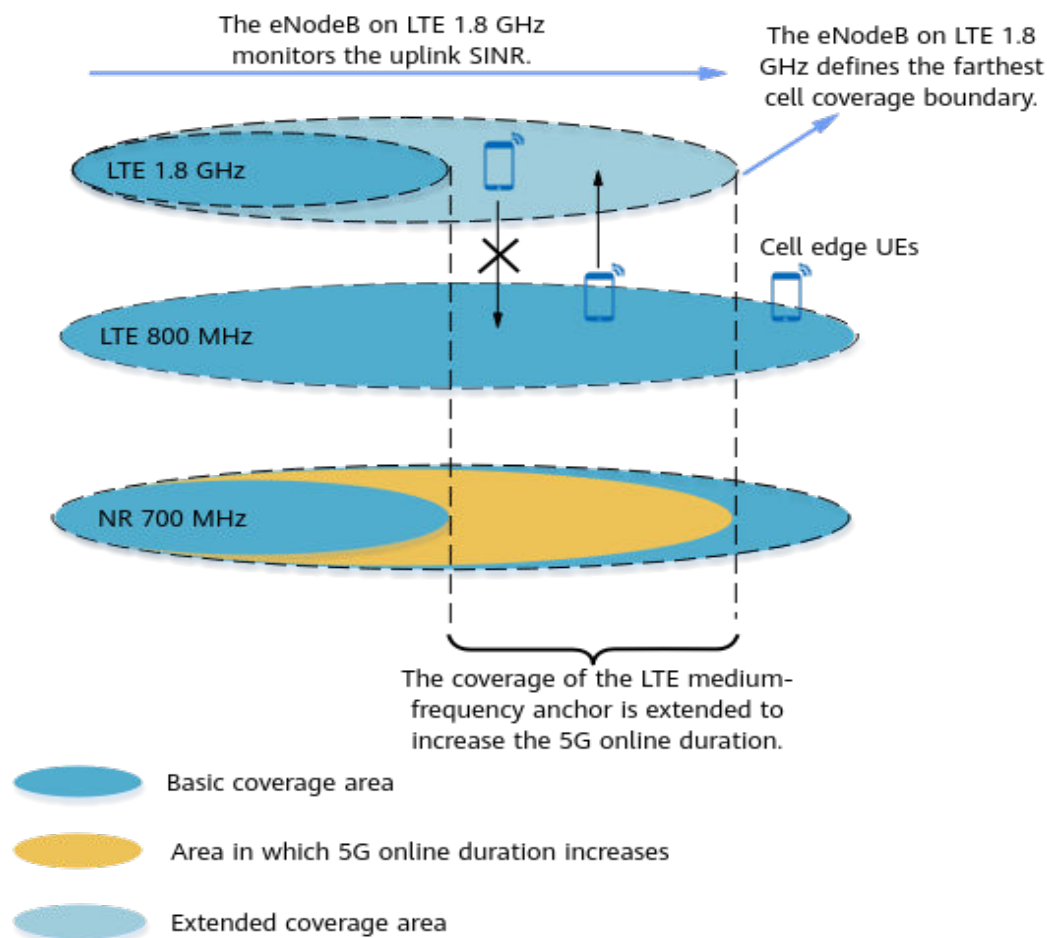
This feature takes effect when the **FDD\_ANCHOR\_ADAPT\_COV\_THLD\_SW** or **TDD\_ANCHOR\_ADAPT\_COV\_THLD\_SW** option of the **NsaDcAlgoParam.NsaDcAlgoExtSwitch** parameter is selected.

This feature includes two functions: threshold adaptation for inter-frequency handovers based on LTE coverage and threshold adaptation for NSA PCC anchoring based on NR coverage.

- Threshold adaptation for inter-frequency handovers based on LTE coverage delays the handover of NSA UEs from LTE medium-frequency cells to LTE low-frequency cells, increasing the 5G online duration of the UEs. As shown in [Figure 4-1](#), the coverage area of the LTE 1.8 GHz cell is extended to delay the handover of UEs in this area to the LTE 800 MHz cell.

- Threshold adaptation for NSA PCC anchoring based on NR coverage increases the probability of UEs being handed over from LTE low-frequency cells to LTE medium-frequency cells. As such, the probability of adding NR low frequencies as SCG frequencies increases, and the 5G online duration of UEs increases. As shown in **Figure 4-1**, the coverage area of the LTE 1.8 GHz cell is extended to increase the probability of handovers from the LTE 800 MHz cell to the LTE 1.8 GHz cell and the probability of adding the NR 700 MHz cell as an SCG cell for UEs.

**Figure 4-1** NSA anchor coverage extension



### 4.1.1 Threshold Adaptation for Inter-Frequency Handovers Based on LTE Coverage

Threshold adaptation for inter-frequency handovers based on LTE coverage delays the handover of NSA UEs from LTE medium-frequency cells to LTE low-frequency cells, increasing the 5G online duration of the UEs.

Threshold adaptation for inter-frequency handovers based on LTE coverage takes effect when either of the following conditions is met:

- EutranInterNFreq.InterFreqHoEventType** set to **EventA4** or **EventA5**

- **EutranInterNFreq.InterFreqHoEventType** set to **EventA3** and the **LOW\_FREQ\_SCS\_SUPPORT\_A3\_SW** option of **CellMultiCarrUniSch.CellMultiCarrUnifiedSchSw** selected

The eNodeB selects the following target UEs for which coverage can be extended in LTE cells:

- NSA UEs for which an SCG has been added
- Non-emergency-call UEs, non-eMTC UEs, or non-eMBMS UEs
- UEs for which GBR bearers are not set up or non-GBR bearers with QCI 69, 70, or 79 are not set up
- UEs that are not blacklisted for this function (this function does not take effect for blacklisted UEs when the **NSA\_ANCHOR\_COV\_EXT\_FORBID\_SW** option of the **UeCompat.BlacklistControlExtSwitch2** parameter is selected.)
- UEs that do not access the local cell through RRC connection reestablishments (this condition needs to be evaluated only when the **REEST\_NSA\_COV\_EXT\_FORBID\_SW** option of the **NsaDcAlgoParam.NsaDcAlgoExtSwitch** parameter is selected.)

After receiving an event A2 measurement report related to coverage-based inter-frequency handovers from a UE for which coverage is to be extended, the eNodeB continuously monitors the uplink SINR of the UE and takes the following actions based on the uplink SINR and downlink RSRP of the UE to ensure that the UE is served by the carrier that can deliver the optimal user experience:

- If the spectral efficiency corresponding to the uplink SINR of the UE is less than **CellMultiCarrUniSch.MobilAdaptCovUISpctEffThld**, the eNodeB immediately performs a coverage-based inter-frequency handover. For details about coverage-based inter-frequency handovers, see *Mobility Management in Connected Mode* in eRAN feature documentation.
- If the downlink RSRP of the UE is less than or equal to **CellMultiCarrUniSch.AdaptCovLowFreqRsrpThld**, the eNodeB immediately performs a coverage-based inter-frequency handover. This procedure is basically the same as that described in *Mobility Management in Connected Mode* in eRAN feature documentation. The differences are as follows:
  - In this case, the threshold for event A4 or threshold 2 for event A5 related to inter-frequency handovers is specified by **CellMultiCarrUniSch.AdaptCovA4RsrpThld**.
  - When the eNodeB receives an event A4/A5 measurement report from a UE for which coverage is to be extended and learns that the cell with the largest RSRP value reported on a neighboring frequency is served by another base station, the eNodeB further checks the **ADAPT\_COV\_THLD\_FORBID\_FLAG** option of the **EutranExternalCell.AggregationAttribute** parameter for the cell.
    - If this option is selected, the eNodeB checks whether the RSRP of the cell with the strongest signal strength is greater than the A4 threshold or A5 threshold 2 for coverage-based inter-frequency handovers. For details, see *Mobility Management in Connected Mode* in eRAN feature documentation.
      - If the RSRP of the cell is greater than the threshold, handovers to the cell are allowed.

- If the RSRP of the cell is less than or equal to the threshold, handovers to the cell are prohibited.
  - If the option is deselected for the cell, handovers to the cell are allowed.
  - If the option is not configured for the cell, handovers to the cell are allowed.
- If the spectral efficiency corresponding to the uplink SINR of the UE is greater than or equal to **CellMultiCarrUniSch.MobilAdaptCovULSpctEffThld** and the downlink RSRP is greater than **CellMultiCarrUniSch.AdaptCovLowFreqRsrpThld**, the eNodeB immediately performs a coverage-based inter-frequency handover. The procedure is basically the same as that described in *Mobility Management in Connected Mode* in eRAN feature documentation. The differences are as follows:
  - The eNodeB can only deliver measurement configurations for an LTE frequency that works with the existing PSCell frequency (one that has been added for the UE) for EN-DC according to UE capabilities and the eNodeB's NR SCG frequency configuration (**NrScgFreqConfig** MO). If no LTE frequency meets the requirements, the eNodeB does not deliver measurement configurations.
  - The value of the **InterFreqHoGroup.InterFreqHoA4ThdRsrp** parameter is taken as the A4 threshold or A5 threshold 2 in the eNodeB-delivered inter-frequency measurement configurations for the frequencies to be measured.

#### NOTE

Threshold adaptation for inter-frequency handovers based on LTE coverage takes effect only when the same offset is configured for the thresholds related to event A2 in handovers between FDD and TDD. For details about the thresholds related to event A2 in handovers between FDD and TDD, see *Mobility Management in Connected Mode* in eRAN feature documentation.

## 4.1.2 Threshold Adaptation for NSA PCC Anchoring Based on NR Coverage

When NSA PCC anchoring based on NR coverage is enabled by selecting the **NSA\_PCC\_ANCHORING\_SWITCH** option of the **NsaDcMgmtConfig.NsaDcAlgoSwitch** parameter and the **NsaDcMgmtConfig.NsaDcPccAnchoringPolicy** parameter is set to **BASED\_ON\_NR\_COVERAGE**, enabling threshold adaptation for NSA PCC anchoring based on NR coverage can increase the probability of handovers of NSA UEs from LTE low frequencies to LTE medium frequencies. This increases the probability of adding NR low frequencies as SCG frequencies in NR low frequency networking scenarios and further improves the 5G camping proportion. Threshold adaptation for NSA PCC anchoring based on NR coverage takes effect only when all of the following conditions are met:

- The PCC anchoring procedure is not triggered by a coverage-based incoming handover.
- The target PCell and the current PCell are served by the same base station.

- After the target PCell and the corresponding SCG are selected by NSA PCC anchoring based on NR coverage, the eNodeB determines that the current PCell does not support the addition of this SCG.
- UEs for which GBR bearers are not set up or non-GBR bearers with QCI 69, 70, or 79 are not set up
- UEs that are not blacklisted for this function (this function does not take effect for blacklisted UEs when the **NSA\_ANCHOR\_COV\_EXT\_FORBID\_SW** option of the **UeCompat.BlacklistControlExtSwitch2** parameter is selected.)
- UEs that do not access the local cell through RRC connection reestablishments (this condition needs to be evaluated only when the **REEST\_NSA\_COV\_EXT\_FORBID\_SW** option of the **NsaDcAlgoParam.NsaDcAlgoExtSwitch** parameter is selected.)

After this function takes effect, the requirements for the target cell during PCC anchoring for NSA UEs are as follows:

- The estimated spectral efficiency corresponding to the uplink SINR after the UEs are handed over to the target cell is greater than or equal to the value of **CellMultiCarrUniSch.MobilAdaptCovULSpctEffThld** configured for the target cell plus 4 dB.
- The downlink RSRP of the target cell is greater than the sum of **CellMultiCarrUniSch.AdaptCovLowFreqRsrpThld** and **CellMultiCarrUniSch.AdaptCovRsrpThldOffset** configured for the target cell.

In addition, if **CellMultiCarrUniSch.AdaptCovLowFreqRsrpThld** of the target cell cannot be obtained, the PCC anchoring threshold specified by **PccFreqCfg.NsaDcPccA4RsrpThld** is used as the downlink RSRP threshold for the target cell.

#### NOTE

Threshold adaptation for NSA PCC anchoring based on NR coverage takes effect only when the A2 threshold offset is set to the same value for handovers between FDD and TDD. For details about the thresholds related to event A2 in handovers between FDD and TDD, see *Mobility Management in Connected Mode* in eRAN feature documentation.

### 4.1.3 No Coverage Extension for High-Speed UEs

Target UEs with NSA anchor coverage extended are located in relatively weak coverage areas. If such a UE moves at a high speed, the signal attenuation is faster and the handover is likely to fail. When the **HighSpeedUeCovNoExtendSw** option of the **CellHoParaCfg.CellHoAlgoSwitch** parameter is selected, the function of no coverage extension for high-speed UEs takes effect. If a UE moves at a speed greater than or equal to 30 km/h, it will be regarded as a high-speed UE by the function. Threshold adaptation for inter-frequency handovers based on LTE coverage and threshold adaptation for NSA PCC anchoring based on NR coverage do not take effect for high-speed UEs.

## 4.2 Network Analysis

## 4.2.1 Benefits

After this feature takes effect, the 5G camping proportion increases. The larger the value of **L.NsaDc.Capable.5gUser.NoScg.BorderUE.Avg**, the higher the gain.

### NOTE

- The frequency band of a frequency with the highest NSA PCC anchoring priority cannot form EN-DC band combinations with some NR frequency bands on the live network.
- The frequency with the highest NSA PCC anchoring priority has weak coverage areas due to wall penetration loss and interference. That is, there are a large number of EN-DC-capable UEs not performing EN-DC in LTE low-frequency cells.

## 4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

The traffic distributions on frequency bands change. As a result, the values of band-specific performance indicators also change. In frequency bands where the traffic volume increases, the uplink throughput of UEs performing full buffer services decreases slightly. The handover success rate decreases and the service drop rate rises due to the extended cell edge.

## 4.3 Requirements

### 4.3.1 Licenses

On the LTE side:

| RAT     | Feature ID    | Feature Name                                           | Model        | Sales Unit |
|---------|---------------|--------------------------------------------------------|--------------|------------|
| LTE FDD | LNOFD-181303  | NSA Anchor Coverage Extension Based on User Experience | LT1SNSACEB00 | Per eNodeB |
| LTE TDD | TDLOFD-181210 | NSA Anchor Coverage Extension Based on User Experience | LT4SNSAUETDD | Per eNodeB |

On the NR side:

No requirement

 NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists (FDD)*, *License Control Item Lists (CIoT)*, and *License Control Item Lists (TDD)* in eRAN feature documentation.

## 4.3.2 Software

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

### 4.3.2.1 Prerequisite Functions

Prerequisite functions of threshold adaptation for NSA PCC anchoring based on NR coverage

| RAT                      | Function Name                          | Function Switch                                                                                                                                                                            | Reference                                    | Description                                                                                                                               |
|--------------------------|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| LTE<br>FDD<br>LTE<br>TDD | NSA PCC anchoring based on NR coverage | <b>NSA_PCC_ANCHORING_SWITCH</b><br>option of the <b>NsaDcMgmt Config.NsaDcAlgoSwitch</b> parameter and <b>NsaDcMgmt Config.NsaDcPccAnchoring Policy</b> set to <b>BASED_ON_NR_COVERAGE</b> | <i>EPC-based NSA Performance Enhancement</i> | Threshold adaptation for NSA PCC anchoring based on NR coverage takes effect only when NSA PCC anchoring based on NR coverage is enabled. |

### 4.3.2.2 Mutually Exclusive Functions

Mutually exclusive functions of threshold adaptation for inter-frequency handovers based on LTE coverage: none

Mutually exclusive functions of threshold adaptation for NSA PCC anchoring based on NR coverage: none

### 4.3.2.3 Function Impacts

Functions related to threshold adaptation for inter-frequency handovers based on LTE coverage:



| RAT                | Function Name                                              | Function Switch                                                                                                                                         | Reference                                   | Description                                                                                                                                                                                                                                                                                                             |
|--------------------|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LTE FDD<br>LTE TDD | NSA carrier combination selection based on user experience | <b>NSA_LTE_FDD_SEL_OP T_SW</b> or <b>NSA_LTE_TDD_SEL_OP T_SW</b> option of the <b>EnodebAlgoExtSwitch</b> . <b>MultiNetworkingOptionOptSw</b> parameter | <i>EPC-based NSA Experience Improvement</i> | After receiving an event A2 measurement report related to coverage-based inter-frequency handovers from a target UE for coverage extension, the eNodeB immediately triggers NSA carrier combination selection based on user experience. The LTE CA band combination of the target UE for coverage extension may change. |

| RAT                | Function Name                                                         | Function Switch                                                                                                  | Reference                                             | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LTE FDD<br>LTE TDD | Adaptive A2 RSRP threshold adjustment for handovers to low-band cells | <b>LOW_FREQ_EXTEND_A2_ADJ_SW</b> option of the <b>MultiCarrUnifiedSch. MultiCarrierUnified-SchEnSw</b> parameter | <i>Low-Band Booster</i> in eRAN feature documentation | Coverage threshold adaptation in Mid- and Low-Band Coordination and threshold adaptation for inter-frequency handovers based on LTE coverage use the same RSRP threshold for triggering event A2. After adaptive A2 RSRP threshold adjustment for handovers to low-band cells in coverage threshold adaptation takes effect, the optimized A2 RSRP threshold also affects the A2 RSRP threshold for threshold adaptation for inter-frequency handovers based on LTE coverage. |

Functions related to threshold adaptation for NSA PCC anchoring based on NR coverage:

| RAT                | Function Name                                                         | Function Switch                                                                                                  | Reference                                             | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LTE FDD<br>LTE TDD | Adaptive A2 RSRP threshold adjustment for handovers to low-band cells | <b>LOW_FREQ_EXTEND_A2_ADJ_SW</b> option of the <b>MultiCarrUnifiedSch. MultiCarrierUnified-SchEnSw</b> parameter | <i>Low-Band Booster</i> in eRAN feature documentation | Coverage threshold adaptation in Mid- and Low-Band Coordination and threshold adaptation for NSA PCC anchoring based on NR coverage use the same RSRP threshold for triggering event A2. After adaptive A2 RSRP threshold adjustment for handovers to low-band cells in coverage threshold adaptation takes effect, the optimized A2 RSRP threshold also affects the A2 RSRP threshold for threshold adaptation for NSA PCC anchoring based on NR coverage. |

### 4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

#### Base Station Models

On the LTE side, the compatible base stations are as follows:

- 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R. 5900 series base stations must be configured with a BBU model other than the BBU5900C.
- DBS3900 LampSite and DBS5900 LampSite

On the NR side, the compatible base stations are as follows:

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910, and 5900 series base stations must be configured with a BBU model other than the BBU5900C.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

#### Boards

Main control boards: UMPTb and later

Baseband processing units: LBBPd and later.

#### RF Modules

This function does not depend on RF modules.

### 4.3.4 Networking

This feature applies to NSA networking involving LTE medium frequencies and NR low frequencies. For details about other requirements, see *EPC-based NSA Basics*.

### 4.3.5 Others

For details, see *EPC-based NSA Basics*.

## 4.4 Operation and Maintenance

### 4.4.1 Data Configuration

#### 4.4.1.1 Data Preparation

[Table 4-1](#) describes the parameters used for function activation.

**Table 4-1** Parameters used for activation

| Parameter Name                                | Parameter ID                                          | Setting Notes                                                                                                                                                                                                                                                                          |
|-----------------------------------------------|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NSA DC Algorithm Extension Switch             | <b>NsaDcAlgoParam.NsaDcAlgoExtSwitch</b>              | Select the <b>FDD_ANCHOR_ADAPT_COV_THLD_SW</b> option for LTE FDD cells.<br>Select the <b>TDD_ANCHOR_ADAPT_COV_THLD_SW</b> option for LTE TDD cells.                                                                                                                                   |
| Mobility Adapt Cov UL Efficiency Thld         | <b>CellMultiCarrUniSch.MobilAdaptCovULSpctEffThld</b> | The value 2 is recommended.                                                                                                                                                                                                                                                            |
| Adaptive Coverage RSRP Threshold for Low Freq | <b>CellMultiCarrUniSch.AdaptCovLowFreqRsrpThld</b>    | The value -122 is recommended. The value of this parameter must be larger than that of <b>InterRatHoCommGroup.InterRatHoA2ThdRsrp</b> .                                                                                                                                                |
| Aggregation Attribute                         | <b>EutranExternalCell.AggregationAttribute</b>        | Select the <b>ADAPT_COV_THLD_FORBID_FLAG</b> option when NSA anchor coverage extension based on user experience is disabled for the eNodeB of the external E-UTRAN cells.                                                                                                              |
| Cell Multi-Carrier Unified Sch Switch         | <b>CellMultiCarrUniSch.CellMultiCarrUnifiedSchSw</b>  | If the serving cell is a medium-frequency LTE cell and is configured with a neighboring E-UTRAN frequency with the <b>EutranInterNFreq.InterFreqHoEventType</b> parameter set to <b>EventA3</b> , you are advised to select the <b>LOW_FREQ_SCS_SUPPORT_A3_SW</b> option of this cell. |

**Table 4-2** describes the parameters used for function optimization.

**Table 4-2** Parameters used for optimization

| Parameter Name                       | Parameter ID                                  | Setting Notes                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------------------------|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Adaptive Coverage A4 RSRP Threshold  | <b>CellMultiCarrUniSch.AdaptCovA4RsrpThld</b> | <ul style="list-style-type: none"> <li>If the local cell is not a low-band cell, you are advised to set this parameter to the <b>CellMultiCarrUniSch.AdaptCovRsrpThld</b> parameter value of the local cell plus 4.</li> <li>If the local cell is a low-band cell, you are advised to set this parameter to the <b>CellMultiCarrUniSch.AdaptCovLowFreqRsrpThld</b> parameter value of the local cell plus 4.</li> </ul> |
| Handover Algo Switch                 | <b>CellHoParaCfg.CellHoAlgoSwitch</b>         | If the coverage area is along express ways, it is recommended that the <b>HighSpeedUeCovNoExtendSw</b> option of this parameter be selected.                                                                                                                                                                                                                                                                            |
| NSA DC Algorithm Extension Switch    | <b>NsaDcAlgoParam.NsaDcAlgoExtSwitch</b>      | It is recommended that the <b>REEST_NSA_COV_EXT_FORBID_SW</b> option of this parameter be selected.                                                                                                                                                                                                                                                                                                                     |
| Blacklist Control Extension Switch 2 | <b>UeCompat.BlacklistControlExtSwitch2</b>    | It is recommended that the <b>NSA_ANCHOR_COV_EXT_FORBID_SW</b> option of this parameter be selected if there are UEs with compatibility issues on the network.                                                                                                                                                                                                                                                          |

#### 4.4.1.2 Using MML Commands

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

## Activation Command Examples

On the eNodeB side

```
//Enabling NSA anchor coverage extension based on user experience and setting related parameters
MOD NSADCALGOPARAM:NsaDcAlgoExtSwitch=FDD_ANCHOR_ADAPT_COV_THLD_SW-1;
MOD NSADCALGOPARAM:NsaDcAlgoExtSwitch=TDD_ANCHOR_ADAPT_COV_THLD_SW-1;
MOD CELLMULTICARRUNISCH: LocalCellId=0, MobilAdaptCovUISpctEffThld=2,
AdaptCovLowFreqRsrpThld=-122, AdaptCovA4RsrpThld=-111;
MOD NSADCMGMTCONFIG: LocalCellId=0, NsaDcAlgoSwitch=NSA_PCC_ANCHORING_SWITCH-1,
NsaDcPccAnchoringPolicy=BASED_ON_NR_COVERAGE;
//(Optional) Selecting the LOW_FREQ_SCS_SUPPORT_A3_SW option if the serving cell is a medium-
frequency LTE cell and is configured with a neighboring E-UTRAN frequency with the InterFreqHoEventType
parameter set to EventA3
MOD CELLMULTICARRUNISCH: LocalCellId=0,
CellMultiCarrUnifiedSchSw=LOW_FREQ_SCS_SUPPORT_A3_SW-1;
```

## Optimization Command Examples

```
//Selecting the HighSpeedUeCovNoExtendSw option
MOD CELLHOPARACFG: LocalCellId=0, CellHoAlgoSwitch=HighSpeedUeCovNoExtendSw-1;
//Selecting the REEST_NSA_COV_EXT_FORBID_SW option
MOD NSADCALGOPARAM: NsaDcAlgoExtSwitch=REEST_NSA_COV_EXT_FORBID_SW-1;
//(Optional) Turning on the blacklist switch if NSA anchor coverage extension based on user experience is
expected not to take effect for a certain type of UEs
ADD UECOMPAT: Index=0, UeInfoType=UE_CAPABILITY, UeCapIndex=0,
BlacklistControlExtSwitch2=NSA_ANCHOR_COV_EXT_FORBID_SW-1;
```

## Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

On the eNodeB side

```
//Disabling NSA anchor coverage extension based on user experience
MOD
NSADCALGOPARAM:NsaDcAlgoExtSwitch=FDD_ANCHOR_ADAPT_COV_THLD_SW-0&TDD_ANCHOR_ADAPT_COV_THLD_SW-0;
//Deselecting the HighSpeedUeCovNoExtendSw option
MOD CELLHOPARACFG: LocalCellId=0, CellHoAlgoSwitch=HighSpeedUeCovNoExtendSw-0;
//Deselecting the REEST_NSA_COV_EXT_FORBID_SW option
MOD NSADCALGOPARAM: NsaDcAlgoExtSwitch=REEST_NSA_COV_EXT_FORBID_SW-0;
```

### 4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

#### NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

## 4.4.2 Activation Verification

After NSA anchor coverage extension based on user experience takes effect, the values of related counters will change.

- The value of **L.HHO.InterFreq.Coverage.PrepAttOut** decreases.
- The value of **L.NsaDC.PCCAnchor.HHO.PrepAttOut** increases.

- The value of **L.NsaDC.PCCAnchor.HHO.ExecAttOut** increases.
- The value of **L.NsaDC.PCCAnchor.HHO.ExecSuccOut** increases.

### 4.4.3 Network Monitoring

After NSA anchor coverage extension based on user experience takes effect, the 5G camping proportion is used for network monitoring.

5G camping proportion = **L.Traffic.User.NsaDc.PCell.Avg**/  
**L.NsaDc.Capable.5gUser.RRC.Avg**



# 5 Parameters

---

The following hyperlinked EXCEL files of parameter documents match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- eNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.

## NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the product documentation delivered with that version.

### **FAQ 1: How do I find the parameters related to a certain feature from parameter reference?**

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

### **FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?**

- Step 1** Open the EXCEL file of the used reserved parameter list.

**Step 2** On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter, which may be only a bit of a parameter. View its information, including the meaning, values, impacts, and product version in which it is activated for use.

----End

# 6 Counters

---

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- eNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

## NOTE

You can find the EXCEL files of performance counter reference for the software version used on the live network from the product documentation delivered with that version.

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

## **FAQ: How do I find the counters related to a certain feature from performance counter reference?**

**Step 1** Open the EXCEL file of performance counter reference.

**Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.

**Step 3** Click **OK**. All counters related to the feature are displayed.

----End

# 7 Glossary

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For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

# 8 Reference Documents

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- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- *ANR Management* in eRAN feature documentation
- *LTE Spectrum Coordination* in eRAN feature documentation
- *RAN Sharing* in eRAN feature documentation
- *Low-Band Booster* in eRAN feature documentation
- *Multi-carrier Unified Scheduling* in eRAN feature documentation
- *Mobility Management in Connected Mode* in eRAN feature documentation
- *Carrier Aggregation* in eRAN feature documentation
- *SA Mobility Management in Connected Mode* in 5G RAN feature documentation
- *Multi-Frequency Smart Selection* in 5G RAN feature documentation
- *Multi-Operator Sharing* in 5G RAN feature documentation
- *Mobility Load Balancing in Connected Mode* in 5G RAN feature documentation
- *Service-differentiated Experience Guarantee* in 5G RAN feature documentation
- *LTE FDD and NR Dynamic Spectrum Sharing*
- *EPC-based NSA Basics*
- *EPC-based NSA Performance Enhancement*
- *NSA Mobility Management*
- *NSA-SA Selection Based on User Experience*
- *SRAN Networking and Evolution Overview*
- *WSFD-021101 5G NSA (Opt.3) Dual Connectivity Management*
- *License Control Item Lists (FDD)* in eRAN feature documentation
- *License Control Item Lists (CIoT)* in eRAN feature documentation
- *License Control Item Lists (TDD)* in eRAN feature documentation
- *License Control Item Lists* in 5G RAN feature documentation